Introduction to Arithmetic

ΝΙΚΟΜΑΧΟΥ ΓΕΡΑΣΗΝΟΥ ΑΡΙΘΜΗΤΙΚΗ ΕΙΣΑΓΩΓΗ

Nicomachus

translated by Claude AI

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Two books of the Introduction to Arithmetic of Nicomachus of Gerasa, the Pythagorean.

Translated from the Greek text of Richard Hoche (Leipzig, 1866). The numbers in the margin give the book, chapter and section of that edition, the standard reference for this work.

Book I

Chapter 1

1.1.1

The ancients, who were the first to pursue science by method, Pythagoras leading the way, defined philosophy as the love of wisdom, as the name itself indeed shows, whereas all before Pythagoras were called wise by an indiscriminate name, just as a carpenter was, and a cobbler, and a helmsman, and in general anyone skilled in some art or craft. But Pythagoras, restricting the name from all these to the science and apprehension of being, and calling wisdom in the proper sense only the knowledge of the truth that lies in this, with good reason also named the desire and pursuit of this wisdom philosophy, as being a desire for wisdom.

1.1.2

He is more worthy of credit than those who define it otherwise, inasmuch as he makes clear the notion of the name and of the thing proper to it. And this wisdom in turn he defined as the science of the truth in the things that are, holding science to be an unerring and unshakeable apprehension of its subject, and the things that are to be those which continue always in the same state and in the same manner in the universe and never depart from being, not even for a

little. These would be the immaterial things, and those by participation in which each of the other things, which are and are called beings homonymously, is said to be a particular thing, and is so.

1.1.3

For bodily and material things, to be sure, are throughout in continuous flux and change, imitating the nature and the property of the eternal matter and substance that was from the beginning; for that was, whole and through its whole extent, mutable and alterable. But the incorporeal things that are contemplated about it or indeed together with it, such as qualities, quantities, configurations, magnitudes, smallnesses, equalities, relations, activities, dispositions, places, times, all those things, in a word, by which what is in each body is encompassed, exist in themselves unmoved and unchangeable, but accidentally partake of and share in the affections of the underlying body.

1.1.4

It is of such things, then, that wisdom is pre-eminently the science, and accidentally also of the things that partake of them, that is, of bodies.

Chapter 2

1.2.1

Now those things are immaterial and eternal and without end, and it is their nature to continue throughout alike and unvarying, abiding in the same manner in their own essence, and each of them is said to be in the proper sense; but the things in generation and destruction, in growth and diminution, in every kind of change and participation, are seen to be varying continually, and they are said to be, homonymously with those others, in so far as they partake of them, but in their own nature they are not truly beings; for they do not remain in the same state even for the briefest moment, but are always passing over, altering in every manner of way, according to Timaeus in Plato, who says:

1.2.2

‘What is that which always is, and has no coming-to-be, and what is that which comes to be, but never is? The one is to be grasped by intellection together with reason, being always and in the same state; the other in its turn is an object of opinion, by opinion together with irrational sense-perception, coming to be and perishing, but never truly being.’

1.2.3

It is reasonable, then, and most necessary, if we aim at the end that is proper and befitting to man, that is, at living well (and this is accomplished through philosophy alone, and by nothing else; and philosophy for us, as I said, is the desire for wisdom, and wisdom the science of the truth in the things that are, and of the things that are some are so called properly, others homonymously), to distinguish with precision and to set out in articulate order the attributes of the things that are.

1.2.4

Of the things that are, then, both those properly so called and those so called by homonymy, that is, both intelligibles and sensibles, some are unified and held together in continuity, as an animal, the universe, a tree, and the like, which are properly and peculiarly called magnitudes; others are divided and set side by side and, as it were, in a heap, which are called multitudes, as a flock, a people, a heap, a chorus, and things of that kind.

1.2.5

Of these two forms, therefore, wisdom must be held to be the science. But since every multitude and every magnitude is of necessity infinite by its own nature (for multitude, beginning from a determinate root, does not cease advancing, and magnitude, being divided from a determinate whole, can nowhere bring the cutting to a stop, but for these reasons goes on to infinity), and since the sciences are in every case sciences of limited things, and never of infinite, it is plain that a science could never be constituted either about magnitude simply or

about multitude simply (for each of them is in itself indeterminate, multitude in the direction of the more, magnitude in the direction of the less), but only about something marked off from each of the two: from multitude, about quantity, and from magnitude, about size.

Chapter 3

1.3.1

Again, to begin afresh: since of quantity one part is viewed by itself, having no relation to anything else, as even, odd, perfect, and the like, and the other as already standing in some manner towards something else and conceived together with its relation to another, as double, greater, less, half, sesquialter, sesquitertian, and the like, it is clear that two scientific methods will lay hold of and make thoroughly plain the whole inquiry concerning quantity: arithmetic that concerning quantity by itself, and music that concerning quantity in relation to another.

1.3.2

And again, since of size one part is in rest and stability, and the other in motion and revolution, two other sciences will in the same way treat accurately of size: geometry of that which abides and is at rest, sphaeric of that which moves and revolves.

1.3.3

Without these, therefore, it is not possible to treat accurately of the forms of being, nor, then, to discover the truth in the things that are, the science of which is wisdom; and it is plain that it is not possible even to philosophize rightly. ‘For what painting contributes to the mechanical arts towards correctness of theory, that same aid do lines and numbers and harmonic intervals and the revolutions of circles give towards the learning of wise doctrines’, says Androcydes the Pythagorean.

1.3.4

And Archytas of Tarentum too, at the beginning of his work on harmonics, says the same thing in some such way as this: ‘They seem

to me to have discerned well concerning the mathematical sciences,[1] and it is in no way strange that they should think rightly about each thing, of what sort it is. For having discerned well concerning the nature of the whole, they were bound to see well also concerning particular things, of what sort they are. So concerning geometry and arithmetic and sphaeric they have handed down to us a clear discernment, and not least concerning music also.

1.3.6

'For these mathematical sciences seem to be sisters; for they are occupied with the two very first forms of being, which are sisters.' And Plato too, at the end of the thirteenth book of the Laws, which some entitle the Philosopher, because in it he considers and determines what manner of man the true philosopher ought to be, in summing up what has been previously discussed and established at greater length, adds: 'Every diagram, and system of number, and every composition of harmony, and the proportion of the revolution of the stars, must be revealed as one to him who learns in the proper manner; and what we say will be seen to be right, if a man learns all things looking to one; for one bond of all these will be revealed. But if anyone is to take philosophy in hand in some other way, he must call on fortune to help him. For without these there is never a road; but this is the manner, these are the sciences, be they hard or easy; by this way one must go, and one must not neglect it. And him who has grasped all these things in this way, as I say, him I call wisest, and I maintain it both in jest and in earnest.' For it is clear that these mathematical sciences are like ladders and bridges, carrying our understanding across from sensible things and objects of opinion to intelligible things and objects of science, and from the material and bodily things that are reared with us and familiar to us from infancy to things unfamiliar and alien to the senses, yet by their immateriality and eternity more akin to our souls, and far more so to the intellective part in them.

1.3.7

So too the Socrates of Plato's Republic, when his interlocutor seems to be bringing forward certain plausible reasons in favour of the mathematical sciences, that they are serviceable for human life, arithmetic for reckonings and distributions and contributions and exchanges and partnerships, geometry for the pitching of camps and the founding of cities and temples and the division of land, music for festivals and merrymakings and the worship of the gods, and sphaeric and astronomy for husbandry and navigation and the other beginnings of undertakings, by showing beforehand the favourable and fitting occasions, says in rebuke: 'How amusing you are, in that you seem to be afraid lest I should be prescribing these sciences as useless! But that is exceedingly hard, or rather impossible; for the eye of the soul, blinded and buried by other pursuits, is by these alone rekindled and roused again, an eye better worth saving than ten thousand bodily eyes; for by it alone is the truth concerning the universe seen.'

Chapter 4

1.4.1

Which, then, of these four methods must we learn first? Plainly that which by nature exists before them all and is more sovereign, and holds the place of principle and root and, as it were, of mother to the others.

1.4.2

And this is arithmetic, not only because we said that it existed before the others in the thought of the craftsman god, as a kind of cosmic and paradigmatic plan, relying upon which, as upon a preliminary design and archetypal pattern, the artificer of the whole sets in order the products wrought out of matter and makes them attain their proper end, but also because it is by nature prior in birth, inasmuch as it abolishes the rest together with itself, but is not abolished together with them. For example, animal is by nature prior to man; for when animal is abolished man too is abolished, but when man is abolished

animal is not then abolished with him. And again man is prior in birth to grammarian; for if there is no man there is no grammarian either, but if there is no grammarian it is possible for man to be. So that, since it abolishes the other along with itself, it is for that reason also the elder.

1.4.3

And conversely that is called younger and later in birth which brings the other in along with itself, but is not brought in along with that other, as for example the musician; for he in every case brings man in along with himself. And again horse; for animal is in every case brought in along with it, but not the reverse; for if animal exists it is not necessary that horse should exist, nor, if man is there, that musician should be brought in with him.

1.4.4

So it is also with the sciences already named. For if geometry exists, arithmetic too must of necessity be brought in along with it; for it is together with arithmetic, and not without the numbers that are brought in along with each, that triangle or square or octahedron or icosahedron or double or octuple or sesquialter, or anything else of the kind that geometry speaks of, can be conceived. For how is it possible for anything to be, or to be called, triple if the number 3 is not presupposed, or octuple if 8 is not presupposed? But conversely 3 and 4 and the rest in order could exist though the figures named after them did not.

1.4.5

Arithmetic, therefore, abolishes geometry along with itself, but is not abolished along with it by geometry; and it is brought in along with geometry, but does not bring geometry in along with itself.

Chapter 5

1.5.1

And again in the case of music: not only because that which is by itself is prior in birth to that which is relative to another, as the great to the

greater and the rich to the richer and man to father, but also because the musical concords, the fourth, the fifth, the octave, are named after number; and likewise they have their harmonic ratios altogether arithmetical, the fourth sesquitertian, the fifth sesquialter, the octave double, the octave and fifth together triple, and the most perfect, the double octave, quadruple.

1.5.2

More evidently still does sphaeric obtain through arithmetic all the inquiries that belong to it, not only because it is later in birth than geometry (for motion is by nature after rest), nor because the movements of the stars have a harmony that is in every way melodious, but also because risings and settings and progressions and retrogradations and occultations and phases of every kind are articulated by the periods and quantities of numbers.

1.5.3

Since, then, it is by nature prior in birth and more honourable and elder, as it were a mother and nurse, we have done well to set its systematic treatment first; and the beginning of that treatment, for the sake of clearness, we shall make from this point.

Chapter 6

1.6.1

All things that have been arranged by nature in the universe according to an artful procedure, both part by part and as wholes, are seen to have been distinguished and set in order in accordance with number by providence and by the intellect that fashioned the whole, the pattern being made fast by the fact that number, pre-existing in the thought of the world-making god, holds as it were the place of a preliminary sketch, number intelligible only and altogether immaterial, yet the true and the eternal essence, so that with reference to it, as to an artistic plan, all these things might be brought to completion: time, motion, heaven, stars, revolutions of every kind.

1.6.2

It is necessary, therefore, that scientific number, since it is set over such things as these, should already be harmonized in accordance with itself, and not by another but by itself.

1.6.3

Now everything that is harmonized is in every case harmonized out of contraries, and out of things that are; for neither can things that are not be harmonized, nor things that are but are like one another, nor things that differ but have no ratio to one another. It remains, then, that the things out of which a harmony is made both are, and are different, and have a ratio to one another.

1.6.4

Of such things, therefore, scientific number also is made; for the very first forms in it are two, having the essence of quantity, and differing from one another, and yet not of different genera, odd and even, and they have been fitted to one another alternately by a wonderful and divine nature, inseparably and in the form of unity, as we shall presently know.

Chapter 7

1.7.1

Number is a determinate multitude, or a combination of units, or a flow of quantity composed of units; and the first division of number is into the even and the odd.

1.7.2

The even is that which can be divided into two equal parts without a unit falling in between; the odd is that which cannot be parted into two equal parts, because of the aforesaid intervention of the unit.

1.7.3

This definition, then, is drawn from the popular conception. But according to the Pythagorean doctrine the even number is that which admits of division, by one and the same act, into the greatest and the least parts, greatest in size and least in quantity, in accordance with

the natural reciprocal relation of these two genera; and the odd is that which cannot have this done to it, but is cut into two unequal parts.

1.7.4

In another manner, according to the ancient account, the even is that which can be cut both into two equal parts and into two unequal, save that the dyad, which is the elementary form in it, admits only one of the two kinds of bisection, that into equal parts; and in any division whatever it shows forth only one of the two species of number, however it be divided in two, without share in the other. The odd is a number which, in any division whatever, a division that is in every case into unequal parts, shows both the two species of number together, never unmixed with one another, but always in one another's company.

1.7.5

And in the definition of each by means of the other, the odd is that which differs by a unit in either direction from an even number, that is, towards the greater and towards the less, and the even is that which differs by a unit in either direction from an odd number, that is, is greater by a unit and less by a unit.

Chapter 8

1.8.1

Every number is half of the numbers on either side of it added together, and is likewise half of those lying next but one on either side, and further of those beyond these, and so on as far as is possible.

1.8.2

The unit alone, because it does not have two numbers on either side of it, is half of one only, the number adjacent to it; the unit, therefore, is the natural principle of all.

1.8.3

By subdivision of the even there is the even-times even, the odd-times even, and the even-times odd. The even-times even and the even-

times odd are opposite to each other, like extremes, while the odd-times even is common to both, like a mean.

1.8.4

The even-times even number, then, is one that can itself be divided into two equal parts, in accordance with the nature of its genus, and that has either of its parts likewise divisible in two, and again in the same way either of the parts of that part divisible into two equal parts, and so on until the division of the successive subdivisions arrives at the unit, which is by nature indivisible.

1.8.5

Take 64, by way of example: the half of this is 32, and of this 16, and the half of this is 8, and of this 4, and of this 2, and then, last of all, the unit is the half of this, which, being by nature indivisible, no longer admits of a half.

1.8.6

It is a consequence for this number that whatever part of it is taken is in every case even-times even in its designation, and that the same part is also even-times even in value, by the quantity of the units in it, and that neither of these two ever shares in any other genus.

1.8.7

Perhaps, then, it has been named even-times even for this very reason, that, being itself even, it has its parts and the parts of its parts, down to the unit, always even both in name and in value; or, to put it otherwise, every part that it has is even-times even in name, and the same part is also even-times even in value.

1.8.8

There is a generation of the even-times even such that none escapes, but all, from a single starting-point, fall under it, and it would come about as follows:[2]

1.8.9

As one advances from the unit, as from a root, in the double ratio to infinity, as many numbers as arise are all even-times even, and it is impossible to find others besides these; for instance, by way of

example, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, and so on to any extent whatever.

1.8.10

Each of the numbers set out, then, has arisen according to the double ratio continued from the unit, and is in every case even-times even; and every part that it is found to have is in every case named from one of the numbers within it, and the combination of units[3] in that part is as great as some one of the numbers within it, yet by a reciprocal arrangement and interchange. If the terms set out by the doublings from the unit are even in number, no single mean can be found, but always two, starting from which the reciprocal arrangement and interchange of parts with values and of values with parts will advance in order, first to the two on either side, then to those lying beyond them on each side, until it reaches the outermost terms, so that the whole corresponds in name to the unit and the unit to the whole. For instance, if we set down 128 as the greatest, the terms set out for it will be even in number, for there are eight in all up to it, and they will not have a single mean, for that is impossible in an even number, but necessarily two, 8 and 16, which will answer to each other in turn; for of the whole, 128, 16 is the eighth part, and conversely 8 is the sixteenth. And as we go on in either direction, 32 is the fourth part and 4 the thirty-second, and again 64 is the half and 2 the sixty-fourth, and last of all, at the extremes, the unit is the one-hundred-and-twenty-eighth part, and conversely 128, corresponding to the unit, is the whole.

1.8.11

But if the series is set out in an odd number of terms, for example in seven, when we have taken 64, the mean will necessarily be one, in accordance with the nature of the odd, and it will answer to itself, because it has no partner, while those on either side of it will always answer to each other, until the correspondence ends at the extremes.

1.8.12

For example, the unit will be the sixty-fourth part, and 64, corresponding to the unit, the whole; and 32 is the half, and 2 the thirty-second; and 16 is the fourth, and 4 the sixteenth; and the eighth part, with nothing set over against it, is 8 itself. It is a property of all the terms set out that, when they are added together cumulatively, they are equal to the term next after them less a unit, so that their sum, however it is taken, will necessarily be an odd number; for the number that is equal to an even number less a unit is always odd.

1.8.13

This observation will be of use to us very shortly for the construction of the perfect numbers. By way of example, the terms within 256, down to the unit, when summed are equal to it less a single unit; and all the terms within 128, the term immediately below it, are likewise equal to it less a single unit; and so always, in the same way, are the terms within each of the successive terms, just as the unit itself is equal, less a unit, to the term after it, that is, to 2, and the two together, less a unit, to the term after them, and the three together, less a unit, to the next; and you will find this going on to infinity without fail.

1.8.14

This too it is most necessary to bear in mind: if the terms set out for the even-times even number that has been taken are even in number, the product of the extremes multiplied by each other will in every case be equal to the product of the mean terms multiplied by each other; but if they are odd in number, the product of the extremes will be equal to the product of the mean term multiplied by itself. For, with an even number of terms, once 128 is equal to eight times 16, and also to twice 64, and again to four times 32, and so throughout; and with an odd number of terms, once 64 is equal to twice 32, and this to four times 16, and this again to eight times 8, the mean term alone being multiplied by itself.

Chapter 9

1.9.1

The even-times odd number is one that is itself also even in genus, but is opposed in species to the even-times even described above; it admits of division into two equal parts, in accordance with the common genus, but has each of its parts at once incapable of being cut into two equal parts, such as 6, 10, 14, 18, 22, 26 and the like; for after each of these has been halved, its parts are at once found to be incapable of halving.

1.9.2

It is a property of this number that every part which it is found to have is opposite in name to its value, and that the quantity of every part is opposite in value to its name, and that never in any way is the value of the part of the same genus as its name. To take the single case of 18: its half, which has an even designation, is 9, odd in value, and its third, conversely, which is named with an odd designation, is 6, even in value; and, by conversion, the sixth is 3 and the ninth 2; and in the other numbers the same manner will be found.

1.9.3

Perhaps, then, it has received such a designation for this very reason, that, though it is even, its halves prove at once to be odd.

1.9.4

This number too is generated when the numbers that differ by a dyad beginning from the unit,[4] that is, the odd numbers, are set out in regular order as far as you wish and are multiplied by a dyad; for the numbers produced would be, in order, these: 6, 10, 14, 18, 22, 26, 30, and so on as far as you care to go. The greater always differ by a tetrad from the lesser numbers next to them. The cause of this is that their original gnomons, that is, the odd numbers, which exceed one another by a dyad, were multiplied by a dyad in order that these might arise, and a dyad multiplying a dyad makes a tetrad.

1.9.5

In the natural series of number, then, the even-times odd numbers will be found fifth from one another, exceeding by a tetrad, passing over three terms, and generated when the odd numbers are multiplied by a dyad.

1.9.6

They are said to be affected in the opposite way to the even-times even numbers, because of these only the greatest extreme is divisible, whereas of those only the least was indivisible; and moreover because in those the reciprocal arrangement of the parts, from the extremes to the mean term or mean terms, made the product of the extremes equal to the square of the mean term or to the product of the means,[5] whereas in these, by the same interchange and examination, the mean term is half of the two extremes added together, or, if the mean terms are two, they too are both together equal to the two extremes.

Chapter 10

1.10.1

The odd-times even number is the one that displays the third species of the even, being common to both of those described, as though it were itself a single mean between two extremes; for it is like the even-times even in one respect and like the even-times odd in another, and in that wherein it departs from the one it shares with the other, and in that wherein it has something in common with the one it differs from the other.

1.10.2

It occurs when an even number that can be divided into two equal parts has its own parts likewise divisible, and sometimes the parts of its parts as well, but cannot carry the resolution of its parts as far as the unit; such are 24, 28, 40; for each of these has a half of its own and in every case a half of its half; and sometimes there is found among them one that admits of the halving being carried still further into its parts; none whatever, however, will have its parts divisible into halves as far as the unit, which is by nature indivisible.

1.10.3

In admitting more than one division, then, it resembles the even-times even and departs from the even-times odd; but in that its divisions never end in the unit, it resembles the even-times odd and departs from the even-times even.

1.10.4

To it alone there belong at once the attributes that belong peculiarly to each of those, and again attributes that belong to neither. For of those the one had only its greatest part divisible, and the other only its smallest indivisible, but this has neither; for it is seen to have more divisions than one in the greater part, and more indivisibles than one in the lesser.

1.10.5

And again there are in it some parts that are not opposite in name to their values nor of a different genus from them, after the likeness of the even-times even, and there are in every case others also that are named oppositely to their values, in a different genus, after the likeness of the even-times odd. In 24, for example, there are parts not opposite in name to their values, the fourth, 6, the half, 12, the sixth, 4, the twelfth, 2; while the third, 8, the eighth, 3, and the twenty-fourth, 1, are affected in the opposite way; and similarly in the rest.

1.10.6

This number is generated by a somewhat more intricate method, showing in a way even in its generation that it is a mixture of both. For since the even-times even arises from the even numbers that proceed by doubling from the unit without end, and the even-times odd from the odd numbers that advance from the triad without end, this number must be woven together out of both genera, as being common to both.

1.10.7

Let us set out, then, the odd numbers from the triad by themselves, in regular order, in one row, 3, 5, 7, 9, 11, 13, 15, 17, 19, and so on, and the even-times even numbers from the tetrad, again in succession, in

another row, according to their own order, 4, 8, 16, 32, 64, 128, 256, and so on, as far as you wish.

1.10.8

Now with the number that stands first in whichever row you please (for it makes no difference) multiply, from the beginning, all those in the other row in succession, and note down the results; then again multiply the same numbers from the top by the second number of the same row, as far as you are able, and write down the products; then again the same numbers from the top by the third number; and however far you go on, no others will be generated for you except the odd-times even numbers.

1.10.9

By way of example, let us use the first number of the row of the odd numbers and multiply by it all those in the other row in order, three times 4, three times 8, three times 16, three times 32, and so on to infinity; for the results will be 12, 24, 48, 96, which must be noted down in one row. Then, from a fresh beginning, do the same again with the second number, five times 4, five times 8, five times 16, five times 32; for these will be produced: 20, 40, 80, 160. Then do the same again with the third number, 7, seven times 4, seven times 8, seven times 16, seven times 32; for the products are 28, 56, 112, 224; and in the same way, as far as you wish to go on, you will find it agree.

1.10.9

The gnomons of the odd-times even

3	5	7	9	11	13	15
4	8	16	32	64	128	256
12	24	48	96	192	384	768
20	40	80	160	320	640	1280
28	56	112	224	448	896	1792
36	72	144	288	576	1152	2304
44	88	176	352	704	1408	2816

The odd-times even numbers
Breadth
Length

1.10.10

Now when you arrange the products from each number in a row of their own, making the rows parallel, there will appear to you, in a wonderful way, along the breadth the peculiar property of the even-times odd, that the mean term is always half of the extremes, if there is one, and, if there are two mean terms, they are equal to the extremes when added together; and along the length the property of the even-times even, for the product of the extremes is equal to the square of the mean term, if there is one, or to the product of the means, if there are two. Thus the peculiar properties of both belong to this number alone, since it is a natural mixture of them.

Chapter 11

1.11.1

The odd, in its turn, is distinguished from the even and has nothing in common with it, since the latter is divisible into two equal parts and the former is indivisible into two equal parts; and of it too, by subdivision, three species are likewise found, differing from one another, of which one is called prime and incomposite, the one opposed to this secondary and composite, and the one that is viewed as lying midway between both of these, like a mean between extremes, is that which is in itself secondary and composite, but relatively to another prime and incomposite.

1.11.2

The very first species, then, the prime and incomposite, occurs when an odd number admits of no other part except the one named after itself, which will also of necessity be the unit; for example, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31. Each of these will by no device be found to have a part with a different name, but only the one named after itself, and this in each of them is always the unit; for 3 has only a third, and 5

only a fifth, and 7 only a seventh, and 11 only an eleventh, and in all of them these parts are the unit.

1.11.3

It has received this name because it can be measured only by the number that is common to all and first of all, the unit, and by no other; and moreover it has not been produced by any other number added to itself, but by the unit alone, 5 by the unit added five times, 7 by the unit added seven times, and the rest according to their own quantity. When these themselves, however, are added to themselves, other numbers can arise, taking their start from these very numbers as from a spring and root, and for this reason they are called prime, as being principles presupposed by those others. And every principle is elementary and incomposite, that into which all things are resolved and out of which all things are composed, while it is itself resolved into nothing and composed out of nothing.

Chapter 12

1.12.1

The secondary and composite number is odd, because it has been distinguished out of one and the same genus, but it has nothing of the nature of a principle in itself; for it took its generation from the adding together of some other number. Hence it is its attribute to possess, besides the part named after itself, a part or parts with a different name as well; the part named after itself is in every case the unit, as in all numbers, but the part or parts with a different name are never the unit, but always that number or those numbers by the adding together of which it was produced, for example, 9, 15, 21, 25, 27, 33, 35, 39. For each of these is measured by the unit, as the others are, and has a part named after itself, as those also have, owing to the nature of the common genus; but, differently from them and more peculiarly, it employs also a part or parts with a different name: 9, besides the ninth, a third as well; 15 a third and a fifth as well, besides the fifteenth; 21 a seventh and a third, besides the twenty-first; and

25, besides the twenty-fifth, which is named after itself, employs also one with a different name, the fifth.

1.12.2

It is called secondary, then, because it can employ yet another measure along with the unit, and because it is not of the nature of a principle, but was itself produced when another number was added to itself or combined with another, 9 from 3, and 15 from 5 or, by Zeus, from 3, and those that follow in the same manner. And it is called composite for this reason, that it can be resolved into those numbers out of which it was composed, since it can also be measured by them; and nothing that can be resolved is incomposite, but is in every case composite.

Chapter 13

1.13.1

While these two species of the odd are opposed to each other, a third is observed between them, fashioned as it were out of both, namely that which in itself is secondary and composite but relatively to another is prime and incomposite. This occurs when a number, in addition to the common measure, the unit, is measured by some other measure as well, and is on that account able to admit a part or parts with a different name besides the part named after itself, yet, when examined against some other number similarly constituted, is found neither able to be measured by a measure common with that number nor to have any part bearing the same name as any whatever of those in that number. For example, 9 compared with 25: for each by itself is secondary and composite, but relatively to each other they have the unit alone as common measure, and no part bears the same name in both, but the third that is in the one is not in the other, nor is the fifth that is in the latter found in the former.

1.13.2

The generation of these numbers is called by Eratosthenes a sieve, because we take the odd numbers mingled together and

undistinguished, and out of them, by this method of generation, we separate, as though by some instrument or sieve, and find the prime and incomposite by themselves, the secondary and composite by themselves, and the mixed apart.

1.13.3

The manner of the sieve is as follows. Setting out all the odd numbers in succession from the triad, in as long a row as possible, I begin with the first and examine which numbers it is able to measure, and I find that it is able to measure those that pass over two intermediate terms, as far as we may wish to proceed; and that it measures them not at random and haphazard, but the one lying first, that is, the one that passes over two intermediate terms, it will measure according to the quantity of the very first number lying in the row, that is, according to its own, for it measures it three times; the one that leaves an interval of two from that, according to the quantity of the number placed second, for it measures it five times; the one further on, again leaving an interval of two, according to the quantity of the number placed third, for it measures it seven times; and the one still further on, lying beyond two more, according to the quantity of the number placed fourth, for it measures it nine times; and so on without limit in the same manner.

1.13.4

Then after this number, making a fresh beginning, I come to the second and examine which numbers it is able to measure, and I find that it measures all those that leave an interval of a tetrad; but the first according to the quantity of the number placed first in the row, for it measures it three times; the second according to that of the second, for it measures it five times; the third according to that of the third, for it measures it seven times; and so on in succession always.

1.13.5

And again, starting afresh, the third number, 7, taking over the measuring, will measure those that leave an interval of six; but the very first according to the quantity of 3, which lies first; the second

according to that of 5, for this is placed second; and the third according to that of 7, for this too has the third position in the row.

1.13.6

And in the same proportion throughout this will go forward for you without hindrance, so that the numbers will succeed to the measuring according to the order in which they lie in the row; the number of terms they pass over will be according to the orderly advance of the even numbers from the dyad to infinity, or according to the doubling of the place in which the measuring number is stationed; and the number of times they measure will be according to the orderly progression of the odd numbers from the triad. If, then, you mark the numbers with certain signs, you will find that those which take over the measuring in turn never all measure the same number at once, and sometimes not even two measure the same number, nor do absolutely all the numbers set out fall under some one of them as measure, but some escape altogether being measured by any whatsoever, some are measured by one only, and some by two or even more.

1.13.8

Those, then, that have in no way been measured but have escaped it are prime and incomposite, sifted out as though by a sieve; those measured by one number only, according to its own quantity,[6] will have only one part with a different name besides the part named after themselves; and those measured by one number, but according to the quantity of another and not its own, or measured by two together, will have several parts with a different name besides the part named after themselves. These, then, will be secondary and composite.

1.13.9

The third division, that which is common to both, which in itself is secondary and composite but relatively to another is prime and incomposite, will consist of the numbers produced when some prime and incomposite number has measured according to its own quantity, if one so produced be compared with another that has its generation

in the same way. For instance 9 (it arose from 3 measuring according to its own quantity, for it measured three times), if it be compared with 25 (it arose from 5 measuring according to its own quantity, for it measured five times): these have no common measure except the unit.

1.13.10

And in order that we may also have a method of discerning those that are, relatively to one another, either prime and incomposite or secondary and composite, seeing that of the former the unit is the common measure, while of the latter there is, besides the unit, some other number as well, and of discerning what this number is:

1.13.11

If two odd numbers should be specified for us, someone propounding them and bidding us discern whether they are prime to one another and incomposite or secondary and composite, and, if secondary and composite, what number is their common measure, one must compare the proposed numbers with each other and always subtract the less from the greater as many times as possible, and then, when this has been subtracted, subtract in turn from the other, again as many times as possible; for this alternation and reciprocal subtraction will of necessity end either in the unit or in some one and the same number, and that necessarily an odd one.

1.13.12

Whenever, then, the subtractions terminate in the unit, they show the numbers to be prime and incomposite relatively to one another; but whenever they terminate in some other number, odd in quantity and occurring twice, say that they are secondary and composite relatively to one another, and that their common measure is that very number which occurs twice. For example, if 23 and 45 were proposed to us, subtract 23 from 45, and 22 will be left; subtracting this in turn from 23, the unit remains; subtracting this from 22 as many times as possible, you will end in the unit. For this reason they are prime and incomposite relatively to one another, and their common measure is the unit that was left.

1.13.13

But if someone should propose other numbers, 21 and 49, I subtract the less from the greater: 28 is left. Then again from this I subtract the same 21 (for it is possible): 7 is left. This I subtract in turn from 21: 14 remains. From this I again subtract 7 (for it is possible): 7 will be left; but a heptad cannot be subtracted from a heptad. Their termination, therefore, has ended in 7 occurring twice, and you are to declare the original numbers, 21 and 49, secondary and composite relatively to one another, and their common measure, besides the universal unit, to be 7.

Chapter 14

1.14.1

Again, to start afresh: of the numbers that are simply even some are superabundant and others deficient, like extremes opposed to each other, and others are midway between the two, and these are called perfect.

1.14.2

And those said to be opposed to each other, the superabundant and the deficient, are divided, in the relation of inequality, into the more and the less; for no other manner of inequality besides these could be conceived, just as neither vice nor disease nor disproportion nor unseemliness nor any such thing could be. For in the more arise excesses and overreachings and extravagances and superfluities, and in the less wants and deficiencies and privations and insufficiencies; but in that which lies between the more and the less, namely the equal, are virtues and healths and moderations and seemliness and beauties and the like, of which the aforesaid species of number, the perfect, is the most generic.

1.14.3

A superabundant number, then, is one which has, over and above the parts that belong and fall to it, others in excess. Just as if some animal were made complete with too many parts or limbs, having ‘ten

tongues', as the poet says, 'and ten mouths', or nine lips, or furnished with three rows of teeth, or hundred-handed, or having too many fingers on one of its hands, so too, if any number, when all the parts in it have been examined and summed up into one total, should be found on comparison to have its own parts greater than itself, this is called superabundant; for it oversteps the symmetry of the perfect number with its own parts. Such are 12, 24 and certain others; for 12 has a half, 6, a third, 4, a fourth, 3, a sixth, 2, and a twelfth, 1, which summed up together make 16, which is greater than the original 12; its parts, therefore, are greater than the whole.

1.14.4

And 24 too has a half, a third, a fourth, a sixth, an eighth, a twelfth and a twenty-fourth, which are 12, 8, 6, 4, 3, 2, 1; and when summed up they amount to 36, which, compared with the original 24, is found to be greater than it, although it is composed of that number's parts alone; here too, therefore, the parts are greater than the whole.

Chapter 15

1.15.1

A deficient number is one which has the property opposite to what has been shown, and has its own parts, when added together into one, less than itself in comparison; as if some animal should fall short of its natural limbs or parts, or if someone should be one-eyed, as in the line 'and a single round eye was set in his forehead', or if someone should be one-handed, or should have fewer than five fingers on one of his hands, or be tongueless, or deprived of some such thing; such a one would be called deficient and, as it were, maimed, after the peculiar character of the number which has its own parts less than itself. Such are 8 and 14; for 8 has as parts a half, a fourth and an eighth, which are 4, 2, 1, and summed up together they come to 7, less than the original number; the parts, therefore, fall short of the completion of the whole.

1.15.2

Again, 14 has a half, a seventh and a fourteenth, which are 7, 2, 1, and all together 10, less than the original number; this number too, therefore, falls short in its parts of having the whole made up out of them.

Chapter 16

1.16.1

While these two species are opposed in the manner, as it were, of extremes, the number called perfect appears as a mean, being found in equality, neither making its own parts, when added together, more than itself, nor showing itself greater than its parts, but always being equal to its own parts. And the equal is in every case viewed as lying on the boundary between the more and the less, and is, as it were, the moderate between the excessive and the deficient, and that which is in unison between the higher and the lower pitch.

1.16.2

Whenever, then, a number, having all the parts that can be in it gathered and summed up and set in comparison with itself, neither exceeds them in multitude nor is exceeded by them, then such a number is properly called perfect, being equal to its own parts. For example, 6 and 28; for 6 has as parts a half, a third and a sixth, which are 3, 2, 1, and these, summed up together and becoming 6, are equal to the original number, and neither more nor less. And 28 has as parts a half, a fourth, a seventh, a fourteenth and a twenty-eighth, which come to 14, 7, 4, 2, 1, and gathered into one they make up 28; and thus neither are the parts more than the whole nor the whole than the parts, but the comparison is in equality, which is the property of the perfect number.

1.16.3

And it has come about that, just as fair things and those in accordance with virtue are rare and easily numbered, while base things and those in vice are abundant, so too the superabundant and the deficient numbers are found to be very many and disorderly, their discovery

being without order, whereas the perfect are easily numbered and arranged in fitting order. For only one is found among the units, 6; only one other among the tens, 28; a third, alone in the rank of the hundreds, 496; and a fourth within the limit of the thousands, that is, below ten thousand, 8128. And it is their attendant property to end alternately in a hexad or an ogdoad, and to be in every case even.

1.16.4

Their production, elegant and sure, neither passing over any of the perfect numbers nor failing to discriminate any of those that are not such, is effected in this manner. You must set out the even-times even numbers from the unit, advancing in succession in a single row, as far as you please, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, then heap them up continually by the addition of one at a time, and at each addition examine the resulting number to see what sort it is. If you find it to be prime and incomposite, you will multiply it by the quantity of the last number taken in, and the product will in every case be perfect; but if it is secondary and composite, you will not multiply, but will add the next number and again consider what the result is; and if it is secondary and composite, you will again pass it by and not multiply, but will add the next; but if it is prime and incomposite, you will multiply it by the last number taken into the sum, and the result will be perfect; and so on to infinity. In like manner you will generate all the perfect numbers in succession, none being passed over. For example, to 1 I add 2 and examine the sum to see what number it is, and I find it to be the number 3, which, by what was demonstrated before, is prime and incomposite; for it has no part of a different name, but only the part named after itself. Therefore I multiply it by the quantity of the number last taken into the sum, that is, 2, and 6 is generated for me; and this I declare to be the first perfect number in actuality, and to have those parts which are seen in the numbers out of which it was composed. For it will have the unit from the part named after itself, that is, the sixth, and 3 from the half,

which is seen in accordance with 2, and conversely the dyad from the third.

1.16.5

And 28 likewise is generated by the same procedure when another number, 4, is added to the former ones; for the sum of the three, 1, 2 and 4, comes to 7, and it is found to be prime and incomposite, for it admits only the part named after itself, the seventh. Therefore I multiply it by the quantity of the last number taken into the sum, and 28 results for me, equal to its own parts, and having itself also the parts in it derived from the numbers already mentioned: the half by reason of the dyad, the fourth by reason of the heptad, the seventh by reason of 4, the fourteenth as the counterpart of the half, and the twenty-eighth by reason of its own name, which in all numbers is the unit.

1.16.6

These having been found, 6 among the units and 28 among the tens, you must do the same for the fashioning of the next.

1.16.7

For again add the next number, 8, and together they come to 15. Examining this, I find it no longer prime and incomposite, but besides the part named after itself it has also a fifth and a third, of different names. Therefore I do not multiply it by 8, but add the next number, 16, and 31 results. Since this is prime and incomposite, it will of necessity be multiplied, according to the general rule of the procedure, by the last number taken into the sum, 16, and 496 will result, among the hundreds; then in the same manner 8128 also, among the thousands; and so on always, as far as one has the strength to follow.

1.16.8

The unit, then, is perfect potentially, but not yet in actuality; for taking it from the row as the very first into the sum, I examined, according to the rule, of what sort it is, and I found it prime and

incomposite; for truly, and not by participation like the others, it is both first of all number and alone incomposite.

1.16.9

I multiply it, then, by the last number taken into the sum, that is, by itself, and the unit is generated for me; for once 1 is a unit.

1.16.10

The unit, therefore, is potentially perfect; for it is equal to its own parts potentially, while the others are so in actuality.

Chapter 17

1.17.1

Now that we have first given the systematic account of quantity by itself, we pass on at this point to relative quantity as well.

1.17.2

Of relative quantity, then, the highest generic divisions are two, equality and inequality; for everything that is viewed in comparison with another thing is either equal or unequal, and there is no third thing besides these.

1.17.3

The equal, then, is observed when of the things compared the one neither exceeds nor falls short when set beside the other, as for example a hundred compared with a hundred, or ten with ten, or two with two, or a mina with a mina, or a talent with a talent, or a cubit with a cubit, and the like, whether in bulk or in length or in weight or in any quantity whatsoever.

1.17.4

This relation, moreover, is in a peculiar way unsplit in itself and indivisible, as being most of the nature of a principle, for it admits of no difference; for of the equal there is not one kind of this sort and another of that, but the equal is so in one and the same manner.

1.17.5

Indeed, that which answers to the equal does not bear a different name from it, but the same name; like friend, neighbour, fellow-

soldier, so also is equal, for the equal is equal to an equal.

1.17.6

The unequal, for its part, is split in two by subdivision, and one part of it is the greater and the other the less, which are opposed in name and contrary to one another in their quantity and relation; for the greater is greater than some other thing, and the less in turn is less than some other thing in comparison, and the names they bear are not the same, but each of the two has a different one, as father and son, striker and struck, teacher and learner, and the like.

1.17.7

Now the greater is divided by a second subdivision into five species, and one of them is the multiple, another the superparticular, another the superpartient, another the multiple superparticular, and another the multiple superpartient.

1.17.8

And of the opposite of this, that is, of the less, there likewise arise by subdivision five species, opposed to the aforesaid five species of the greater (as whole to whole, the less to the greater, so also each to each in the order already stated, being distinguished in opposition by the prefix sub-): the submultiple, the subsuperparticular, the subsuperpartient, the submultiple superparticular, and the submultiple superpartient.

Chapter 18

1.18.1

To begin, then, from the top, the multiple is the species of the greater that is the very first and by nature the earlier in origin, as we shall presently see, and it is a number which, when it is viewed in comparison with another, contains the whole of that number in itself more than once. For example, all the successive numbers beginning with the dyad, when compared with the unit, generate in their proper sequence the orderly species of the multiple; for 2, first, both is and is called the double, 3 the triple, 4 the quadruple, and so on without

limit; for ‘more than once’ signifies either twice or thrice or so on in succession, as far as you please.

1.18.2

Set over against this is the submultiple, which is itself by nature the very first in the lesser part of inequality, and it is a number which, when it is compared with a greater, is able to measure it completely more than once; and ‘more than once’ begins from twice and goes on without limit.

1.18.3

If, then, it measures the greater number in the comparison only twice, it is called specifically the subdouble, as 1 is of 2; if thrice, the subtriple, as 1 is of 3; if four times, the subquadruple, as the same 1 is of 4, and so on in succession.

1.18.4

And since each of them, both the multiple and the submultiple, is generically unlimited, the differences by subdivision and the species also are observed, further, to proceed by nature without limit; for the double, beginning from 2, proceeds through all the even numbers, as we take every other number out of the natural flow; and these will be called doubles in comparison with the even and odd numbers set down in succession from the unit.

1.18.5

Triples are all those which, from the beginning, are picked out as third in order, two being passed over, as 3, 6, 9, 12, 15, 18, 21, 24, and it is an attribute of these that they are alternately even and odd, and they too are triples of all the successive numbers in the orderly series of number from the unit, going on as far as one wishes to follow.

1.18.6

Quadruples are those which are in every case fourth, three being passed over, as 4, 8, 12, 16, 20, 24, 28, 32 and so on in succession, and these too are quadruples of the numbers in regular order from the unit, going on as far as one has the strength to follow. It is an attribute of these also that they are all even; for one need only pass over every

other one of those same even numbers that were distinguished above, so that it necessarily belongs to the numbers that are simply even that all of them are doubles, every other one a quadruple, one after every two a sextuple, and one after every three an octuple, and so the advance goes on analogously without limit.

1.18.7

Quintuples will be seen to be those which pass over four and are ranged fifth from one another, and are themselves quintuples of the numbers ranged in succession from the unit, and alternately one is odd and one even, after the same order as the triples.

Chapter 19

1.19.1

The superparticular, the second species of the greater both in nature and in order, is a number which contains in itself the whole of the number compared with it and some one part of it besides.

1.19.2

But if the part is a half, the greater of the numbers compared is called specifically sesquialter, and the less subsesquialter; and if it is a third, sesquitertian and subsesquitertian; and you will find it always agreeing thus as you go forward throughout, so that these species also proceed without limit, although they are species of a genus that is itself unlimited. For the very first of them, the sesquialter, turns out to have as its consequents the even numbers in succession from the dyad, and no other whatever, and as its antecedents the triples in succession from the triad, and no other.

1.19.3

These must be coupled in regular order, first with first, second with second, third with third, 3 with 2, 6 with 4, 9 with 6, 12 with 8, and the corresponding numbers with those of the same rank.

1.19.4

If we wish to examine the second species of the superparticular, that is, the sesquitertian (for the part that naturally comes next after the

half is the third), we shall have this as its definition: a number which contains in itself both the whole of the number compared with it and a third part of it in addition to the whole. As examples of it in regular order we shall take the successive quadruples from the tetrad coupled with the triples from the triad, those of the same rank with those of the same rank, as 4 with 3, 8 with 6, 12 with 9, and in the same way as far as one pleases.

1.19.5

It is clear that the number which answers to the sesquitertian, and is called with the prefix sub- the subsesquitertian, is the one that is contained in the former, both the whole of it and a third of itself besides, as 3 in 4, 6 in 8, 9 in 12, and those that follow among the numbers of the same rank.

1.19.6

We must observe the elegant feature that attends all these, that the first ones, which are also called the roots, are next to one another in the natural flow, while the second from the root leave out only one number, the third two, the fourth three, the fifth four, and so on always, as far as you please.

1.19.7

And further this also, that the part after which each of the superparticulars is named is observed in the lesser of the roots, and never in the greater.

1.19.8

That the multiple is by nature, and not by our ordinance, more primitive and older than the superparticular, we shall very soon learn in what follows, in a more elaborate way; but here too, for a simple illustration, we must set out in orderly and parallel rows the multiples already described by us, species by species, first the double in one row, then in a second the triple, then the quadruple in a third, and so on up to the decuple, in order that we may perceive their order and their variety and their methodical advance, and which is prior by nature, and moreover certain other delightful and elegant consequences.

1.19.9

Let the diagram be as follows:[7]

1.19.9

Length

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

1.19.10

Let there be set out in the first row the natural series of number from the unit, and then in succession those of the species of the multiple that have been prescribed.

1.19.11

Now then, of the first rows, which begin from the unit and run in breadth and in depth in the form of the letter gamma, the second rows in either direction, themselves also in the form of a gamma and beginning from the tetrad, are multiples according to the first species of the multiple, for they are doubles; and the first differs from the first by a unit, the second from the second by a dyad, the third from the third by a triad, the next by a tetrad, those after them by a pentad, and you will find this following consistently throughout. The third rows in either direction, beginning from the ennead[8] that is common to them, will be triples of the numbers in the same first row according to the second species of the multiple, the cross-lines running to the triad on either side being examined along with them.

1.19.12

The difference too will advance along with these according to the nature of the even numbers, being for the first a dyad, for the next a tetrad, for the third a hexad; and this difference nature has of her own accord interposed for us between these rows under examination, as is evident in the diagram.

1.19.13

The fourth row, of which the common beginning in either direction is 16, while its cross-lines come to an end at the tetrads, shows the third species of the multiple, that is, the quadruple, when its numbers are examined against the same very first row, rank for rank, the first number against the first, the second against the second, the third against the third, and so on; and again the differences of these are a triad, a hexad, then an ennead, then a dodecad, and the quantities that advance by the triad; and these too are detected in the fabric of the diagram, set out in order just above the quadruples themselves; and in the species of the multiple that follow the analogous thing goes on throughout.

1.19.14

In comparison with the second row in either direction, which begins from the common beginning 4 and runs out crosswise to the dyad proper to each side, the rows next below it in order display the very first species of the superparticular, that is, the sesquialter, numbers of the same rank being compared with numbers of the same rank; thus by divine nature, and not by our convention or agreement, the superparticulars are of later origin than the multiples; for example, 3 of 2, 6 of 4, 9 of 6, 12 of 8, 15 of 10, and so on throughout; and these too have as their difference the successive numbers from the unit, like those before them.

1.19.15

The sesquitertians, the second species of the superparticular, go forward with an equal and similar advance, starting from 4 compared with 3, and 8 with 6, and 12 with 9, and 16 with 12, and correspondingly take on an equal increase of their differences as well.

1.19.16

And in the remaining relations both of the multiple and of the superparticular you will see, as you go forward without limit, that the results are in agreement and in no way contradictory.

1.19.17

This also has been wrought in the diagram with no less exactness: at its corners there are units, the one at the beginning a simple unit, the one at the end a unit of the third order (the hundred), and the two that remain units of the second order (the tens), occurring twice, so as to make the product equal to the square.

1.19.18

Moreover the advance from the unit to the decads is equal on either side, and so again, on the opposite sides, are both the progressions from the decad to the hundred.

1.19.19

And the numbers on the diagonal from the unit to the hundred are all squares, the products of equals multiplied by equals, while those that flank them on either side are all heteromecic, unequal, and the products of sides greater one than the other by a unit; so that from two successive squares taken once and the heteromecic number between them taken twice a square is in every case produced, and conversely from two adjacent heteromecic numbers taken once and the square between them taken twice a square is in every case produced.

1.19.20

And many other such delightful things anyone who was zealous might find showing themselves in this diagram, about which it is not now the time to speak at length; for we have not yet gained the knowledge of them from the Introduction, so that we must turn to what comes next. For after these two generic relations of the multiple and the superparticular, and the other two opposed to them and expressed with the prefix sub-, the submultiple and the subsuperparticular,

there is in the greater part of the unequal the superpartient, and in the lesser its opposite, the subsuperpartient.

Chapter 20

1.20.1

It is the superpartient relation when a number contains in itself the whole of the number compared with it and in addition more than one part of it; and 'more than one' begins once again from 2 and proceeds to all the numbers in succession; so that the root of the superpartient is, reasonably, the one that has two parts of the number compared with it in addition to the whole, and it will be called specifically superbipartient; after this the one that has three in addition to the whole will be called specifically supertripartient, and after this the superquadripartient, then the superquintipartient, and so on always.

1.20.2

The parts have their root and beginning from the third; for it is impossible here to begin from the half. For if we should suppose some number to have 2 halves of the opposite number in addition to the whole, we shall without noticing it be positing a multiple instead of a superpartient; for every whole together with 2 halves of itself becomes the double of the original. So it is most necessary to begin from 2 thirds, then 2 fifths, then 2 sevenths, and after these 2 ninths, following the progression of the odd numbers; for 2 fourths, for instance, are again a half, and 2 sixths a third, and so superparticulars will once more arise instead of superpartients, which is not what is set before us, nor is it in keeping with the consistency of the systematic treatment.

1.20.3

After the superpartient the subsuperpartient at once arises along with it, when a number is itself contained whole in the number compared with it, and in addition several parts of it, either 2 or 3 or 4 or 5 and so on.

Chapter 21

1.21.1

The order of both and their generation in due sequence are found when we set out the even and odd numbers in succession from the triad and compare with these the pure odd numbers alone, in unbroken succession from the pentad, first with first, as 5 with 3, and second with second, as 7 with 4, and third with third, as 9 with 5, and fourth with fourth, as 11 with 6, and so on in the same order as far as one pleases. For in this way the species of the superpartient and of the subsuperpartient will be shown in regular order in the roots of each, the superbipartient first, then the supertripartient and the superquadripartient and the superquintipartient, and so on further in like manner; for after the roots of each the succeeding numbers will arise when both the terms are doubled or tripled and, in general, magnified according to the species of the multiple in their regular order.

1.21.1

The roots

5 3	7 4	9 5	11 6	13 7
10 6	14 8	18 10	22 12	26 14
15 9	21 12	27 15	33 18	39 21
20 12	28 16	36 20	44 24	52 28
25 15	35 20	45 25	55 30	65 35
30 18	42 24	54 30	66 36	78 42
35 21	49 28	63 35	77 42	91 49
40 24	56 32	72 40	88 48	104 56
45 27	63 36	81 45	99 54	117 63
superbipartients of thirds	supertripartients of fourths	superquadripartients of fifths	superquintipartients of sixths	supersextipartient of sevenths

1.21.2

It must be noticed that from the two parts which are contained in the greater number in addition to the whole the third is understood, in the case of the three the fourth, in the case of the four the fifth, in the

case of the five the sixth, and so on always, so that the progression in the naming is of some such kind as this: superbipartient of thirds, supertripartient of fourths, superquadripartient of fifths, then superquintipartient of sixths, and similarly in the rest.

1.21.3

The simple and uncompounded relations of relative quantity, then, are these that have been described; those compounded out of them and, as it were, woven together from two into one are the following, of which the antecedents are the multiple superparticular and the multiple superpartient, and the consequents those which at once arise along with each of these, named with the prefix sub-: with the multiple superparticular the submultiple superparticular, and with the multiple superpartient the submultiple superpartient; and in the subdivision of the genera the specific relations will answer to the specific, these too being named with the prefix sub-.

Chapter 22

1.22.1

It is the multiple superparticular relation, then, when the greater of the terms compared contains within itself the lesser more than once and, in addition to this, some one part of it, of whatever kind.

1.22.2

Being, as it were, composite, such a number is diversified in two ways, according to the proper character of each of the names combined in it; for since the multiple superparticular is generically compounded of the multiple and the superparticular, it will have in its specific subdivisions a certain diversity and variation, separately in the former part of its name and separately in the second: as, in the former, that of the multiple, the double or triple or quadruple or quintuple and so on in succession; and in the second, derived from the superparticular as genus, its species in their regular order, the sesquialter, the sesquitertian, the sesquiquartan, the sesquiquintan and so on in succession, so that the combination advances in some such order as

this: double sesquialter, double sesquitertian, double sesquiquartan, double sesquiquintan, double sesquisextan, and so on analogously; and from another beginning triple sesquialter, triple sesquitertian, triple sesquiquartan, triple sesquiquintan; and again from the beginning quadruple sesquialter, quadruple sesquitertian, quadruple sesquiquartan, quadruple sesquiquintan; and again from the beginning quintuple sesquialter, quintuple sesquitertian, quintuple sesquiquartan, quintuple sesquiquintan, and those that correspond to these without limit. For as many times as the greater contains the whole of the lesser within itself, from that quantity the former part of the ratio, of those that are combined in the multiple superparticular, will take its name; and of whatever kind the part is which, in addition to the whole taken several times, is present in the greater, after that will be named the second species of ratio from which the multiple superparticular is compounded.

1.22.3

Examples of it: 5 is the double sesquialter of 2, 7 the double sesquitertian of 3, 9 the double sesquiquartan of 4, 11 the double sesquiquintan of 5; and you will always generate them thus in regular order by comparing with the even and odd numbers in succession from the dyad the pure odd numbers from the pentad, first with first, second with second, third with third, and the rest, those of like rank with those of like rank; and of all the even numbers in succession from the dyad, the consecutive numbers from the pentad that differ by a pentad will be pure double sesquialters, those of like rank of those of like rank; and if, beginning from three,[9] all the numbers that differ by a triad are set out, as 3, 6, 9, 12, 15, 18, 21, and in another row those that begin from the heptad and differ by a heptad are set out without limit, as 7, 14, 21, 28, 35, 42, 49, and the greater are compared with the lesser, first with first, second with second, third with third, fourth with fourth, and so on, the second species, that of the double sesquitertians, comes to light, set out in its proper regular order.

1.22.4

Then again, from another beginning, if the pure row of the quadruples is set out, 4, 8, 12, 16, 20, 24, 28, 32, and then beside it in another row are set out the consecutive numbers that begin from the ennead and advance by steps of an ennead, as 9, 18, 27, 36, 45, 54,[10] we shall again have the specific multiple superparticular coming to light, that is, the double sesquiquartan, in regular order; and anyone who wishes may conceive this carried on without limit.

1.22.5

The other species begins with the triple sesquialter, such as 7 to 2 and 14 to 4 and, in general, the numbers that advance by a heptad compared with the even numbers in regular order from the dyad.

1.22.6

Then again, starting afresh, 10 to 3 is the first triple sesquitertian and 20 to 6 the second triple sesquitertian, and in general the decuples in succession compared with the triples in succession. These things we can observe more exactly and more clearly in the diagram with which we are already acquainted; for in relation to the first row those that lie next in order, whole row compared with whole row, display the species of the multiple in regular order without limit, all of them being compared always with the same first row; and when each single row is compared with all those above it in succession, each with its neighbour, the beginning being made by us from the second row, all the species of the superparticular are generated in their proper regular order; and beginning from the third row, all the rows of odd rank from the fifth on, when set over against it first and against those continuous with it, one by one, will display all the species of the superpartient in regular order; and the comparisons of the multiple superparticular will have a natural order of their own if, beginning from the second row, we compare with it the numbers of the fifth, first with first and second with second and third with third and so on in succession, and with the third row those of the seventh, and with the fourth those of the ninth, and so on according to the fitting regular order, as far as one has the strength to follow.

1.22.7

It is plain that here too the lesser terms are named in correspondence to the greater with the prefix sub-, according to the designations assigned to them all.

Chapter 23

1.23.1

The multiple superpartient is the remaining relation of number. This, and the relation named in correspondence to it with the prefix sub-, exists when a number contains within itself the whole of the number compared with it more than once (that is, twice or thrice or any number of times whatever) and, in addition to this, certain parts of it, more than one, either 2 or 3 or 4 or 5 and so on in succession.

1.23.2

These parts are not halves, for the reasons already stated, but either thirds or fourths or fifths, and so on in the like sequence.

1.23.3

From what has been said before it is not difficult to understand the species of this relation also, how they are varied in the same way as those before it and without any difference: double superbipartient, then double supertripartient, then double superquadripartient, and so on analogously; for example, 8 is the double superbipartient of 3 and 16 the double superbipartient of 6, and universally the numbers that begin from the ogdoad and differ by an ogdoad are so of those that begin from the triad and differ by a triad, those of like rank of those of like rank; and in the case of the remaining species one could find the regular order by following what has been said before. Here too it must be understood that the corresponding name of the term compared goes forward with the prefix sub- and changes along with the other.

1.23.4

And thus the ten arithmetical relations reach for us the end of their study, as in a first introduction. There is, however, a more elegant method, and one most necessary for the whole study of the nature of

the universe, which shows us most clearly and indisputably that, first, the fair and the definite and that which falls under science is by nature prior to the indefinite and incomprehensible and ugly; and next, that the parts and species of the unlimited and indefinite are given shape and bounded by the former, and attain their fitting order and regular arrangement, and that, as though by some seal or measure, all the things that fall under it come to share in its likeness and its name. For in this way it is reasonable that the rational part of the soul also will be that which sets in order the irrational, and that spirit and appetite, which are ranked in the two species of inequality, will be brought into regular order by the reasoning part as by a kind of equality and sameness.

1.23.5

And from this equalizing there will rightly result for us the so-called ethical virtues, temperance, courage, gentleness, self-control, endurance and the like.

1.23.6

Come then, let us consider of what sort is the theorem that bears upon these matters of nature. It is one that demonstrates that from equality, sole and very first, as from a kind of mother and root, are generated all the various species of inequality and the differences of the species.

1.23.7

Let there be set before us certain equal numbers in three terms, first units, then dyads in three other terms, then triads, and next tetrads, after that pentads, and so on as far as you wish. For when these have been set out in this way, by a design that is divine and not human, nay, that comes from nature herself, multiples will be produced first, and of these the double will lead, after it the triple, upon this the quadruple, then the quintuple, and so on without limit in the order with which we are already acquainted; second, the superparticular, and of this again the very first species, the sesquialter, will lead, upon it the one after it, the sesquitertian, and upon these the next in order, the sesquiquartan and sesquiquintan and sesquisextan, and so on

analogously without limit; third, the superpartient, and again of this the superbipartient will lead, and directly upon it follows the supertripartient, then the superquadripartient, and directly the superquintipartient, and so on, as far as one may advance, in accordance with what has gone before.

1.23.8

You must have, then, certain rules, like natural laws unswerving and inviolable, by which the whole aforesaid advance and progress from equality will go forward prosperously without deserting its order. These rules are: make the first equal to the first, the second equal to the first and second together, and the third equal to the first, two seconds and the third together. For if you fashion according to these laws, you will get at once all the species of the multiple in order, sprouting, as it were, and growing forth out of the three terms of equality set out, without your contriving anything or lending any help; and from equality directly the double, from the double directly the triple, from the triple next the quadruple, and from this the quintuple, in regular order, and so on for ever.

1.23.9

From these same multiples in their regular order, when they are reversed, there are at once generated by a kind of natural necessity, through the same three rules, the superparticulars, and these not at random nor without order but in the proper sequence: from the first, the double, reversed, the first, the sesquialter; from the second, the triple, the second among them, the sesquitertian; then the sesquiquartan from the quadruple, and in general each from that after which it is named.

1.23.10

And from another beginning, when the superparticulars themselves are set out just as they were produced, but in reverse order, there are generated the superpartients, which by nature come after them: from the sesquialter the superbipartient, from the sesquitertian the

supertripartient, and the superquadripartient from the sesquiquartan, and so on without limit in the same proportion.

1.23.11

But when the superparticulars in their regular order are not reversed but set out directly, there are generated through the same rules the multiple superparticulars: the double sesquialter from the first, the sesquialter; the double sesquitertian from the second, the sesquitertian; the double sesquiquartan from the third, the sesquiquartan; and so on always.

1.23.12

And from those generated by the reversal of the superparticulars, that is, the superpartients, and from those generated without reversal, that is, the multiple superparticulars, there are again produced in the same way through the same rules, when they are set out directly and when they are reversed, the numbers that exhibit the remaining relations.

1.23.13

Of all that has been said, of their generation and their order, and of the direct and the reversed arrangement, let the following examples suffice us by way of reminder.

1.23.14

From the relation and proportion in sesquialters, when it is reversed so as to begin from the greater term, there is constituted a relation in superpartient ratios, the superbipartient of thirds; and when it is set out directly, from the lesser term, a multiple superparticular, namely the double sesquialter; as from 9, 6, 4 either 9, 15, 25 or 4, 10, 25; and from that in sesquitertians, beginning from the greater, a superpartient, namely the supertripartient of fourths, and from the lesser the double sesquitertian; as from 16, 12, 9 either 16, 28, 49 or 9, 21, 49; and from that in sesquiquartans, beginning from the greater, a superpartient, namely the superquadripartient of fifths, and from the lesser a multiple superparticular, namely the double sesquiquartan; as from 25, 20, 16 either 25, 45, 81 or 16, 36, 81.

1.23.15

In all the relations that are thus disjoined, and in that from which both come, the last term, a square, remains the same,[11] while the first changes to the lesser; and in every case the extremes are squares.

1.23.16

Moreover, in another way also the multiple superpartients and superpartients of another kind come to light from the superpartients; for example, from the superbipartient of thirds, beginning from the lesser term, the double superbipartient of thirds, and from the greater the supertripartient of fifths; as from 9, 15, 25 either 9, 24, 64 or 25, 40, 64; and from the supertripartient of fourths, beginning from the smaller, the double supertripartient of fourths, and from the greater the superquadripartient of sevenths; as in the case of 16, 28, 49 either 16, 44, 121 or 49, 77, 121.

1.23.17

And again, from the superquadripartient of fifths, such as 25, 45, 81, beginning from the lesser comes the double superquadripartient of fifths, in the terms 25, 70, 196, and from the greater again a superpartient, the superquintipartient of ninths, as in the terms 81, 126, 196; and in what follows, without limit, you will find things analogous and in harmony.[12]

1.23.17

1. (§ 8) See how from equality the species of the multiple are generated in regular order

–	1	1	1
double	1	2	4
triple	1	3	9
quadruple	1	4	16
quintuple	1	5	25
sextuple	1	6	36
septuple	1	7	49
octuple	1	8	64
nonuple	1	9	81
decuple	1	10	100
doubles reversed	4	2	1
sesquialters	4	6	9
triples reversed	9	3	1
sesquitertians	9	12	16
quadruples	16	4	1
sesquiquartans	16	20	25
25	5	1	
25	30	36	
36	6	1	
36	42	49	
49	7	1	
49	56	64	
64	8	1	
64	72	81	
81	9	1	
81	90	100	
100	10	1	
100	110	121	
sesquialters reversed	sesquitertians	sesquiquartans	
9 6 4	16 12 9	25 20 16	
superbipartients	supertripartients	superquadripartients	
9 15 25	16 28 49	25 45 81	
sesquiquintans	sesquisextans	sesquiseptimans	

36 30 25	49 42 36	64 56 49	
superquintipartients	supersextipartients	superseptipartients	
36 66 121	49 91 169	64 120 225	
sesquioctaves	sesquinonans	sesquidecimans	
81 72 64	100 90 81	121 110 100	
superoctipartients	supernonipartients	superdecipartients	
81 153 289	100 190 361	121 231 441	

2. (§ 9) See how the sesquialters come from the doubles reversed
See how the sesquitertians come from the triples, these too being reversed
See how the sesquiquartans also come from the quadruples reversed
How also from the quintuples come the sesquiquintans
How from the sextuples come the sesquisextans
How from the septuples come the sesquiseptimans
And from the octuples the sesquioctaves
And from the nonuples the sesquinonans
And from the decuples the sesquidecimans
3. (§ 10) See how the superparticulars, set out in reverse order, have generated the superpartients
The species of the superparticular

1.23.17

End of the first book of the Introduction to Arithmetic, in two books, of the Gerasene.

Book II

Chapter 1

2.1.1

Since that is called, and is, an element, out of which as the least thing something is composed and into which as the least it is resolved (as, for instance, letters are called the elements of written speech, for out of them is the composition of all articulate speech and into them, as the ultimate things, it is resolved; and notes are the elements of all

melody, from which it begins to be put together and into which it is resolved; and, in common, of the universe the so-called four elements are simple bodies, fire, water, air, earth, for out of these, as the very first things, the constitution of the whole is explained by the study of nature, and into these, as the last, its resolution is conceived), we wish to prove that equality too is an element of relative quantity; for of quantity absolute and by itself the monad and the dyad were the most primary elements, out of which, as the least things, it is forever composed and increased even without limit, and at which, when it is resolved towards the less, it comes to a stand.

2.1.2

Now the progress and increase in the case of inequality we have shown to proceed from equality to absolutely all the relations, with a certain regular order, by means of three rules; it remains, then, in order that it may truly be an element, to prove that the resolutions also come to their end in it as the ultimate term. A general method of the following kind must be known.

Chapter 2

2.2.1

When three terms are given you in any relation and proportion whatever, whether multiple or superparticular or superpartient or one compounded of these, multiple superparticular or multiple superpartient, provided only that the mean term be seen to stand in the same ratio to the lesser as the greater to the mean, or the reverse, always subtract the lesser from the mean, whether it be set first or last, and put the lesser itself as first term; what is left of the second after the subtraction set down as second term; and when one such first term and two such second terms have been subtracted from the remaining one, that is, from the greater of those given you, make what is left the third term; and the resulting terms will be in some other relation, prior by nature.

2.2.2

And if again from these very terms you take away in the same manner what is left of a term, your three terms will be found to have been carried back into three others nearer to the root; and this you will find always coming about consistently, until they are reduced to equality; whence it is plainly altogether necessary to declare that equality is in every way the element of relative quantity.

2.2.3

There attends upon this study a most elegant theorem, and one most useful both for the Platonic generation of the soul and for all the harmonic intervals; for there we are frequently bidden, for instance, to set off in succession two sesquialter ratios, or three, or four, or five, or on without limit, or two sesquitertians, or sesquiquartans, or sesquioctaves, or superparticulars of whatever kind, and of each of these three, or four, or five, or as many as one may prescribe.

2.2.4

And it is reasonable to do this not in a layman's fashion and unscientifically, and sometimes even mistakenly, but with art and without stumbling and most swiftly, by a method such as the following.

Chapter 3

2.3.1

Every multiple will stand at the head of as many superparticular ratios answering to it in name as is its own place in order from the unit, and of neither more nor fewer by any contrivance whatever.

2.3.2

The doubles, then, will beget sesquialters, the first one, the second two, the third three, the fourth four, the fifth five, the sixth six, and neither more nor fewer; but by every necessity, when the superparticulars begotten have received their due quantity, having become equal in number to the multiples that begot them, at that very point, by some divine contrivance, the number that brings them all to an end is found to be by nature incapable of admitting that part

according to which the superparticulars were advancing. From the triples all the sesquitertians will advance, they too, the begotten, equal in number to those that beget them, and coming to an end, after the sufficiency of their advance, in numbers that do not admit a third; and the sesquiquartans in the same way from the quadruples, receiving as their summit, after their sufficient progression, a number that does not admit a fourth.

2.3.3

For example, by way of illustration, since the doubles beget sesquialters equal to them in number, the upper row, the first, will be that of the multiples, 1, 2, 4, 8, 16, 32, 64. Now since 2 is the first after the unit, it will be the origin of one sesquialter only, 3, which does not admit a half so that another might become its sesquialter; the first double, therefore, is generative of one sesquialter only, and the second, 4, generative of two sesquialters, for 6 is the sesquialter of 4 itself, and 9 of 6, but of 9 there is no other, for it has no half. And 8, being the third double, will be the father of three sesquialters: one, 12, in relation to itself; another, 18, in relation to 12; a third, 27, in relation to 18; but of a fourth no longer, because of the general rule, for 27 no longer admits a half. And 16, being the fourth double, will stand at the head of four sesquialters, 24, 36, 54 and, last, 81, so that they may of necessity be equal in number to those that begot them, for 81 by nature no longer admits a half. And this you will find to hold analogously as you go on without limit.

2.3.4

For example, for the sake of illustration, let a diagram of the double be drawn thus.

2.3.4

Diagram of the doubles

In breadth, double

1	2	4	8	16	32	64
	3	6	12	24	48	96
		9	18	36	72	144
			27	54	108	216
				81	162	324
					243	486
						729

Along the hypotenuse, triple
In depth, sesquialter

Chapter 4

2.4.1

For the triple a similar illustration must be drawn.

2.4.1

Diagram of the triples

In breadth, triple

1	3	9	27	81	243	729
	4	12	36	108	324	972
		16	48	144	432	1296
			64	192	576	1728
				256	768	2304
					1024	3072
						4096

Along the hypotenuse, quadruple
In depth, sesquitertian

2.4.1

In this, in the same way, we shall see the first triple, 3, standing at the head of one sesquitertian ratio only, that of 4 to it, which at once shuts off the generation of another like it; for 4 does not admit a third, and so will not have a sesquitertian either. The second triple is 9; for this

reason it will be the origin of two sesquitertian ratios only, that of 12 to it and that of 16 to 12; and 16 thereupon checks the progression, for it does not admit a third, and therefore neither has it any sesquitertian in itself.

2.4.2

Next in order is placed the triple 27, in the third place from the unit as the triples advance, 1, 3, 9, 27; for this reason it too will be the origin of three sesquitertian ratios only, and by no means of more; for the first, that of 27 itself, is 36; the second, that of 36, is 48; and the third, that of 48, is 64, which no longer has a third part and therefore does not admit a sesquitertian; and the fourth is the leader of four ratios, and the fifth, plainly, of five.

2.4.3

Such is the illustration; and for the remaining multiples too let your manner of drawing the diagrams be the same, as you observe that here also nature, just as we found in what was systematically set forth before, shows us the doubles to be prior to the triples, the triples to the quadruples, these to the quintuples, and so on always throughout. For if the topmost rows in breadth are doubles, they will have those that lie parallel beneath them of the like kind, but the diagonals that subtend them will be the next species of the same genus, greater by a unit, that is, triples, when they are examined in parallel lines; and if those in breadth should be triples, the diagonals will in every case be quadruples; and if the former are quadruples, the latter are at once quintuples; and so on for ever.

Chapter 5

2.5.1

It remains, after making clear what other ratios the combinations of ratios produce, to pass on to what follows next in the Introduction.

2.5.2

Now the first two ratios of the superparticular, taken together into one, beget the first ratio of the multiple, that is, the double; for every

double will be a combination of sesquialter and sesquitertian, and every sesquialter and sesquitertian put together will in every case produce one double. For example, since 3 is the sesquialter of 2, and 4 the sesquitertian of 3, 4 will be the double of 2, being compounded of sesquialter and sesquitertian. Again, since 6 is the double of 3, we shall find ranged between them some number which of necessity preserves the sesquitertian ratio to the one and the sesquialter to the other; 4, at any rate, lying between 6 and 3, yields the sesquitertian ratio to 3 and the sesquialter to 6.

2.5.3

Rightly, therefore, was it said that the double, when it is resolved, is resolved into sesquialter and sesquitertian, and that when sesquialter and sesquitertian are put together, in every case only a double is constituted, and that the two very first species of the superparticular, put together, are productive of the very first species of the multiples.

2.5.4

Again, from another beginning, this very first species of the multiple, thus begotten, together with the first species of the superparticulars, becomes productive of the next species of the same genus, that is, of the second multiple, which is the triple; for from every double and sesquialter put together a triple of necessity springs. For example, since 12 is the double of 6, and 18 the sesquialter of this same 12, 18 is at once also the triple of 6. And in another way, if I choose to make the mean term not 12 but rather 9, the sesquialter of 6, the same result, unvarying and in agreement, will follow for me; for 18, being the double of 9, will preserve the triple ratio to 6. From sesquialter and double, therefore, the first species of superparticular and of multiple, when they are mingled, there is constituted the second species of the multiple, the triple, and into them it is in every case resolved.

2.5.5

For see, 6, being the triple of 2, will have 3 as mean term, which will exhibit two ratios, the sesquialter in relation to 2 and, in relation to itself, the double ratio of 6. And if this triple too, being the second

species of the multiple, be put together with the sesquitertian, which is the second species of the superparticular, there would arise from the two the next species of the multiple, that is, the quadruple, which will also of necessity be resolved into the two, according to the same nature as in the cases already shown; and the quadruple, taking to itself the sesquiquartan, will be productive of the quintuple, and that again, with the sesquiquintan, of the sextuple, and so on throughout, so that the multiples in regular order from the beginning, together with the superparticulars in regular order from the beginning, are found to produce the next multiples in the direction of the greater; for the double with the sesquialter is productive of the triple, the triple with the sesquitertian of the quadruple, the quadruple with the sesquiquartan of the quintuple, and, as far as you care to go on, nothing contrary will be seen to befall you.

Chapter 6

2.6.1

Up to this point, then, we have discoursed sufficiently of quantity that stands in some relation to another, having measured out by selection what is fitting and easily grasped for the capacity of those who are just being introduced; for what remains on this topic will be filled in after we have in turn broken off and first treated systematically certain other matters whose investigation is more to the purpose, drawn from the attributes of quantity by itself and not of that which stands in some relation to another; for the theorems in the mathematical sciences are always wont, somehow, to be articulated and made clear through one another. What must first be surveyed and contemplated beforehand concerns linear numbers and plane and solid, cubic and spherical, equilateral and scalene, bricks and beams and wedges and the like, which are properly taught in the Introduction to Geometry, as being more akin to size, but are taken up besides, in a more seminal way, in arithmetic, as in the mother of that science, more primordial than it; for we remember that a little while

ago it appeared to us to be such as abolishes the other sciences together with itself, but is not abolished together with them, and conversely is of necessity implied along with them, but does not imply them along with itself.

2.6.2

First, however, it must be recognized that each letter by which we denote a number, such as ι, by which we denote ten, κ, twenty, ω, by which eight hundred, signifies the number by human convention and agreement, and not by nature; whereas the natural and unartificial, and for that reason the simplest, notation of numbers would be the setting out side by side of the units contained in each. For example, the writing of one unit by a single alpha will be the sign of one; the setting out of two units side by side, that is, of two alphas, will be the sign of the dyad; of three lying in a straight line with one another, it will be the character of the triad; of four ranged in a straight line, of the tetrad; of five, of the pentad; and so on always. For through such writing and signification alone could the graphic representation of the plane and solid numbers that are to be described be rendered distinct and clear, thus:

2.6.2

Unit	α
Dyad	α α
Triad	α α α
Tetrad	α α α α
Pentad	α α α α α

2.6.2

and so on always to greater numbers, analogously.

2.6.3

The monad, then, occupying the place and the character of a point, will be the principle of intervals and of numbers, but not yet an interval or a number, as the point is the principle of line and of interval, but not yet a line or an interval. Indeed, a point added to a

point makes nothing more, for what is without extension added to what is without extension will have no interval, just as if one were to consider nothing added to nothing, for it makes nothing. For it appeared to us in the same way also in the case of equality among the relations: a proportion is indeed preserved, as the first is to the second, so the second to the third, yet no interval is generated between the extremes in relation to one another, as there is in all the other relations apart from equality. In the very same way the monad also, alone out of all number, when it multiplies itself begets nothing more than itself. The monad, therefore, is without extension and has the nature of a principle, and interval is first found and seen in the dyad, then in the triad, then in the tetrad, and successively in those that follow; for an interval is that which is observed between two terms.

2.6.4

The first dimension is called line, for a line is that which is extended in one direction; two dimensions, surface, for a surface is that which is extended in two directions; three dimensions, solid, for a solid is that which is extended in three directions, and it is in no way possible to conceive a solid that has got more dimensions than three, depth, breadth, length; for by these are defined the six directions which are said to belong to every body, according to which motions in place are distinguished: forward, backward, up, down, right, left; for upon each dimension two directions, opposed to each other, of necessity attend: upon one, up and down; upon another, forward and backward; and upon the third, to the right and to the left.

2.6.5

And the statement, as it happens, may be converted thus: if a thing is solid, it has in every case the three dimensions, length, depth, breadth; and conversely, if a thing has the three dimensions, it is in every case solid, and nothing else.

2.6.6

That which has only two dimensions, therefore, will not be a solid but a surface, for this admits of two dimensions only; and in its case too it is possible to convert the statement in like manner, for a surface is rightly that which is extended in two directions, and again that which is extended in two directions will in every case be a surface.

2.6.7

The surface, therefore, is less than the solid by one dimension, and the line less than the surface by one, being that which is extended in one direction and having got one dimension only, and falling short of the solid by two dimensions; and of the line itself the point falls short by one dimension; for this reason it was said before to be without extension, falling short of the solid by three dimensions, of the surface by two, and of the line by one.

Chapter 7

2.7.1

The point, then, is the principle of dimension, but not a dimension, and the same is also the principle of line, but not a line; and the line is the principle of surface, but not a surface, and the principle of that which is extended in two directions, but not itself extended in two directions.

2.7.2

And with good reason the surface is the principle of body, but not a body, and the same is the principle of that which is extended in three directions, but not itself extended in three directions.

2.7.3

So too among numbers the monad is the principle of every number that is advanced unit by unit in one dimension; the linear number is the principle of the plane number, which is broadened out as a plane in a second dimension; and the plane number is the principle of the solid number, which acquires besides, in a third dimension, a certain depth in addition to the original ones. For example, by subdivision, linear numbers are simply all those that begin from the dyad and

advance by the addition of a unit in one and the same dimension; plane numbers are those that begin from the triad as their most original root and proceed through the next successive numbers, taking their designation too according to the same order; for very first come the triangular, then after them the square, then after them the pentagonal, then upon these the hexagonal and heptagonal, and so on without limit; and they are named, as we said, from the numbers that lie in succession from the triad onward.

2.7.4

The triangle, therefore, is found to be the most original and most elementary figure of planes; for in the geometrical planes too, if straight lines are drawn from the angles to the centres, every rectilinear figure will in every case be resolved into as many triangles as it happens to have sides, whereas the triangle itself, undergoing the same treatment as the others, will not pass over into anything else, but into itself. The triangle, therefore, is an element in those also; for all the others are of necessity resolved into it, but it into no other; and out of it the rest would be constituted, but it out of none. This, therefore, is the element of the others, and nothing is the element of it.

2.7.5

And in the arithmetical planes too the argument as it proceeds will confirm what is said.

Chapter 8

2.8.1

A triangular number, then, is one which, when it is resolved into units, gives to the arrangement of its parts in a plane an equilateral figure of triangular form; examples of it are 3, 6, 10, 15, 21, 28 and those that follow in order; for their graphic representations will be regular, at once triangular and equilateral, and as you advance, as far as you please, you will find such a number taking triangular form, if you set before all the rest, as the most elementary, that which arises from the

unit, so that the unit too may appear as potentially a triangle, and 3 as the first in actuality.

2.8.2

Their sides will increase by the successive number; for the side of the one potentially first is a unit, the side of the first in actuality, that is, of 3, is a dyad, the side of the second in actuality, that is, of 6, is a triad, the side of the third is a tetrad, of the fourth a pentad, of the fifth a hexad, and so on always.

2.8.3

It is generated when the natural series of number is set out in a row and the successive numbers, from the beginning, are continually added together one at a time, for with each addition and accumulation the triangles are completed in regular order. For example, from this natural row, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, taking the very first I have the triangle potentially first, the unit; then, heaping upon it the one next to it, I have the first triangle in actuality, for 2 and 1 are 3, and in its graphic representation it is put together thus:[13] beneath one unit two units are set side by side, and the number 3 is made a triangle; then next, 3, the successive number, heaped upon these, spread out into units and added, yields 6, the second triangle in actuality, and gives it its figure besides; and again the number that follows by nature, 4, heaped upon these and written beneath unit by unit, yields the one that comes in regular order after those mentioned, 10, and takes a triangular shape; and 5 upon this, then 6, then 7 and all those that follow, so that, fittingly, the sides of each are also of as many units as the numbers from the natural row that were added together for its composition.

Chapter 9

2.9.1

A square number is the one next in order after this, which shows in its diagram no longer three angles, like the former, but four, though it too is in an equilateral figure, for example 1, 4, 9, 16, 25, 36, 49, 64, 81, 100;

for the diagrams of these turn out to be equilateral squares in some such way as this;[14] and so on in succession, as far as you please.

2.9.2

It is a property of these also, as of those before them, that the progression of their sides advances according to the natural series of number; for the one potentially first, 1, has a unit as its side; the first in actuality, 4, a dyad; the second in actuality, 9, a triad; the one after it, the third in actuality, 16, a tetrad; the fourth a pentad, the fifth a hexad, and so in general, in order, with those that follow.

2.9.3

This number too is generated when the natural series of number is set out in a row and there are heaped upon the unit no longer the successive numbers one after another, as has been shown, but all those that lie with one omitted between them, that is, the odd numbers; for the first, 1, is potentially the first square; the second, 1 and 3, is the first square in actuality; the third, 1 and 3 and 5, is the second square in actuality; the fourth, 1 and 3 and 5 and 7, is the third square in actuality; and the next arises when 9 is heaped upon the former ones, and the one after it when 11 is added, and so on always.

2.9.4

In these too it is the case that the side of each is of as many units as there are numbers heaped together for its generation.

Chapter 10

2.10.1

A pentagonal number is one which likewise, upon being spread out into units, is represented graphically in a plane as a pentagonal figure equilateral on every side, for example 1, 5, 12, 22, 35, 51, 70 and the analogous ones.

2.10.2

But each side of the first in actuality, that is, of 5, is a dyad, for a unit is the side of the pentagon potentially first of all, 1; of 12, the second of those set out, the side is a triad, of the one after it, 22, a tetrad, of the

next, 35, a pentad, and a hexad that of the one following this, 51, and so on always; for in general the side is of as many units as the numbers that were heaped together for its composition, picked out from the arithmetical flow set out in a row in natural order; for in a like and similar manner there are heaped upon one another, for the generation of the pentagon, the numbers which, beginning from the unit, leave two between them, to any extent whatever, that is, those which exceed one another by a triad. The unit is potentially the first, and is represented graphically thus;[15] 5 is the second, put together from 1 and 4, and is itself represented graphically thus; and 12, the third, is from the two former and 7 heaped upon them, so that it too may have a triad as its side, since three were put together for its composition, just as the one before it, 5, had a dyad as its side, being put together from two; and its graphic representation is as follows. Those after these will arise as there are heaped on in succession the numbers that stand in regular order after the heptad at a difference of a triad, for example 10, 13, 16, 19, 22, 25 and so to infinity; for they will be 22, 35, 51, 70, 92, 117, and this without end.

Chapter 11

2.11.1

Hexagons and heptagons and those that follow will be carried forward by the same method from the natural flow of number set out in a row, the intervals always increasing by the addition of a unit; for as the triangle was completed by receiving into its heap the numbers that differ by a unit and omit none, and the square those that differ by a dyad and omit one, and the pentagon accordingly those that differ by a triad and omit two, which we also showed by setting out examples both of them and of the numbers completed from them, so too the hexagons will have as gnomons those that differ by a tetrad and omit three, from which, put together in a heap, they are completed, for example 1, 5, 9, 13, 17, 21 and so on in order, so that the hexagons

completed are 1, 6, 15, 28, 45, 66 and so on always, as far as one may wish.

2.11.2

The heptagons, which follow these, have as their gnomons numbers that differ by a pentad and omit four between them, for example 1, 6, 11, 16, 21, 26, 31, 36 and so to any extent whatever, while the numbers themselves that are composed are 1, 7, 18, 34, 55, 81, 112, 148, and this without end.

2.11.3

The octagons advance in the same order, differing by a hexad in their gnomons, and correspondingly in their sums.

2.11.4

And so that, as you observe it in all cases, you may have this as a universal rule in agreement with them: the gnomons of each polygon differ from one another by a dyad less than the quantity of the angles in its name, that is, by a unit for the triangle, by a dyad for the square, by a triad for the pentagon, by a tetrad for the hexagon, by a pentad for the heptagon, and so on always by regular increase.

Chapter 12

2.12.1

Concerning the nature of the plane polygons, then, this is sufficient for a first introduction. But that the teaching about them is in the fullest harmony with that by lines and not out of tune with it would be clear not only from the graphic representation of each, but also from the following: every square figure divided diagonally is resolved into two triangles, and every square number is resolved into two successive triangular numbers, and is therefore composed of two successive triangular numbers. For example, the triangles are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55 and those that follow, and the squares 1, 4, 9, 16, 25, 36, 49, 64, 81, 100.

2.12.2

If, then, you add together any two successive triangles you please, you will in every case make a square, and therefore, whatever square you resolve, you will be able to make two triangles out of it. And again, a triangle joined to any square figure, on whatever side, makes a pentagon; for example, the triangle 1 joined to the square 4 makes the pentagon 5; and the next one, namely 3, added to the next, 9, makes the pentagon 12; and 6, which follows, added on to 16, which follows, yields 22, which follows; and 10 added to 25 yields 35, and so on always.

2.12.3

In the same way, if the triangles should be added to the pentagons also, in the same order, they will generate the hexagons in regular order, and again the same triangles interwoven with those will make the heptagons in order, and after those the octagons, and this to infinity.

2.12.4

As a reminder, let there be set out for us rows of polygons written in parallel, as follows: the first of triangles, the one after it of squares, after both of these that of pentagons, then of hexagons, then of heptagons, then, if one should wish, of the succeeding polygons as well.

2.12.4

Length and breadth

Triangles	1	3	6	10	15	21	28	36	45	55
Squares	1	4	9	16	25	36	49	64	81	100
Pentagons	1	5	12	22	35	51	70	92	117	145
Hexagons	1	6	15	28	45	66	91	120	153	190
Heptagons	1	7	18	34	55	81	112	148	189	235

Depth

2.12.4

It is possible to make the setting out of the succeeding polygons also in parallel rows in this way.

2.12.5

For in general you will find that the squares are the sum of the triangles of the same rank above them and, further, of those of the same kind that lie next before these, for example 4 of 3 and 1, 9 of 6 and 3, 16 of 10 and 6, 25 of 15 and 10, and 36 of 21 and 15, and so on without end; and that the pentagons are the sum of the squares of the same rank above them and, in addition, of the primal triangles which fall one place short of the same rank, for example 5 of 4 and 1, 12 of 9 and 3, 22 of 16 and 6, 35 of 25 and 10, and so on always.

2.12.6

Again, the hexagons are in like manner the sum of the pentagons of the same rank above them and of the triangles previously set out, for example 6 of 5 and 1, 15 of 12 and 3, 28 of 22 and 6, 45 of 35 and 10, and so on as far as you please.

2.12.7

With the heptagons the manner is the same; for 7 is the sum of 6 and 1, 18 of 15 and 3, 34 of 28 and 6, and those that follow accordingly, so that each polygon is the sum of the polygon above it of the same rank, which has one angle fewer, and of the triangle in the topmost row, the one that lies one place short of the same rank.

2.12.8

With good reason, therefore, the triangle is the element of the polygons both in lines and in numbers; for indeed, both in depth and in breadth in the diagram,[16] the successive numbers along the rows themselves are always found to have as their differences the triangles in regular order.

Chapter 13

2.13.1

From this it is now easy to see both what the solid number is and how such a number advances with equal sides; for the number which, in addition to the two dimensions observed in the plane representation, in length and in breadth, has taken on a third dimension, which some

call depth, others thickness, and some height, that would be a solid number, extended in three directions and having in itself length, depth and breadth.

2.13.2

This number first of all shows itself in what are called pyramids. These arise by tapering from broader bases to a sharp apex, first in triangular fashion from a triangular base, secondly in square fashion from a square base, next after these in pentagonal fashion from a pentagonal base, then analogously from a hexagon and a heptagon and an octagon, and so on always to infinity.

2.13.3

Just so, of course, in the geometrical solid figures too: if from an equilateral triangle one should conceive three straight lines from its angles, equal in length to the sides of the triangle, converging in height to one and the same point, a pyramid would be completed, contained by four triangles, equilateral and equal to one another, one the original triangle and three those marked off by the three straight lines mentioned.

2.13.4

And again, if from a plane square one should think of four straight lines equal in length to the sides of the square, each to each, converging once more in height to one and the same point, a pyramid would be completed from a square base, tapering in square fashion, and contained by four equilateral triangles and one square, the original one.

2.13.5

And from a pentagon and a hexagon and a heptagon, and as far as one wishes to go, in the same way straight lines equal in number to the angles, raised from the angles themselves and converging to one and the same point, bring to an apex a pyramid named from its pentagonal or hexagonal or heptagonal base, or analogously.

2.13.6

So too among numbers, from the unit, as from a point, every linear number has grown, for example 1, 2, 3, 4, 5 and those that follow to infinity; and from these same numbers, which are linear and extended in one dimension, put together in a certain way and not at random, the polygonal and plane numbers are fashioned, triangles when the gnomons are put together with none omitted, squares with one omitted, pentagons with two omitted, and so on always.

2.13.7

In the very same way, when these plane polygonal numbers themselves are heaped upon one another and, as it were, built up, the pyramids of the same kind as each are generated, that from a triangular base from the triangles themselves, that from a square base from the squares themselves, that from a pentagon from the pentagons, and that from a hexagon from the hexagons, and so throughout.

2.13.8

The pyramids from a triangular base, then, in regular order are these: 1, 4, 10, 20, 35, 56, 84 and so on in succession, and their origin is the triangles themselves heaped upon one another, first 1, then 1 and 3, then 1, 3 and 6, then 10 in addition to these, and next 15 together with the foregoing, and upon these 21, and next 28, and so to infinity.

2.13.9

And it is clear that the greater of the numbers is conceived as the lowest, for it is found to be the base, and the one immediately after it as above it, and the one after that as above this, until the unit appears at the apex and, as it were, tapers off the completion of the pyramid into a point.

Chapter 14

2.14.1

The pyramids next in order are those from a square base, rising with the same shape to one and the same point. These are fashioned in the same way as the triangular pyramids already described; for having set

out in a row the squares in regular order from the unit, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 and those that follow, I again place them one upon another in a heap, in depth, 1 on top of 4, and there arises the pyramid 5, the first in actuality from a square base; for here too the unit is the first potentially.

2.14.2

Again, this same pyramid of five units, just as it is, I place entire upon the square 9, and there is constructed for me the pyramid 14 from a square base, having a triad as its side on every side, whereas the one before it, 5, had a dyad, and the one potentially first of all a unit; for here also each side of any pyramid whatever must be of as many units as are, in number, the polygons that have been heaped together for its construction.

2.14.3

For again, the pyramid 14 as a whole, which has the square 9 as its base, I place upon the square 16, and there is completed for me the pyramid 30, which is the third in actuality of those from a square base. And by the same order and procedure we shall construct pyramids from a pentagonal base also, and from a hexagonal and a heptagonal base, and going on continually farther, by heaping upon one another the polygons corresponding to each, beginning from the unit as from the least and going on to infinity in each case.

2.14.4

And from this it becomes clear that triangles are the more elementary; for absolutely all the pyramids that are exhibited and shown, from the polygonal bases of each kind, are contained by triangles as far as the apex.

2.14.5

But that we may not be ignorant of truncated, twice-truncated and thrice-truncated pyramids, whose names we shall meet with in treatises, especially the theoretical ones, it must be known that if a pyramid from any base whatever, that is, one having as its base any polygon at all, whether triangle or square or pentagon or any of the

succeeding polygons of the same kind, when it is increased by heaping does not taper to a unit, it is called simply truncated, being deprived of the natural culmination that belongs to all; for it does not end in the potential polygon, the unit, as in some one point, but in another, an actual one, and a unit is no longer its apex, but its limit proves to be a plane with the same number of angles as the base. But if, in addition to not ending in a unit, it furthermore does not end in the first actual polygon next to the unit, such a pyramid is called twice-truncated; and if, further still, it has not the second actual polygon at its termination, but only the one beneath it, it will be called thrice-truncated, and indeed four-times truncated if it has not the one after that either, and five-times truncated in the next case, and so on continually, as far as you wish to extend the name.

Chapter 15

2.15.1

Such, then, is the generation, advance, increase and nature of the equilateral solid numbers of pyramidal form, having as its seed and root the polygonal numbers themselves and the orderly heaping up of these. But there is another regular ordering, that of solids of a different kind, the so-called cubes, beams, bricks, wedges, spherical numbers and parallelepipedal numbers, which has an order of progression somewhat as follows.

2.15.2

The squares previously mentioned, 1, 4, 9, 16, 25, 36, 49, 64 and those that follow, which are extended in two dimensions and in their plane representation have length and breadth only, will take on a third dimension as well and will be solid and extended in three dimensions, if each is multiplied by its own side: 4, which is twice 2, being again taken twice, so that an ogdoad is produced; 9, which is three times 3, being again increased by a triad in another dimension and becoming 27; 16, which is four times 4, being again enlarged by the tetrad, its

own side, and becoming 64; and those that follow similarly, without end.

2.15.3

Here too the sides will be of as many units as were those of the squares from which they arose, each from each: the sides of 8 will be dyads, as are those of 4; those of 27 triads, as are those of 9; those of 64 tetrads, as are those of 16; and so on in succession; so that the side of the unit also, the potential cube, will be a unit on every side, exactly as is that of the unit as potential square.

2.15.4

In general, each square is one plane, and has four angles and four sides; but each cube, since it is an increase out of each square multiplied by its own side, will in every case have six planes, each of which is equal to its parent square, and twelve sides, each of which is equal to, and indeed of the same number of units as, the side of the parent square, and eight solid angles, each of which is contained by three sides, each such as is each of those in the parent square.

Chapter 16

2.16.1

Since, then, the cube is a solid figure equilateral in every direction, in length, depth and breadth, and equally extended towards the six so-called directions, it follows that there is opposed to it that which has its dimensions nowhere equal to one another, but the dimension of depth unequal to that of breadth, and that of length unequal to each of these, for example two times three times four, or two times four times eight, or three times five times twelve, or according to some other such inequality.

2.16.2

Such solid figures, whose dimensions are everywhere unequal to one another, are called simply scalene; but some, using many names for them, call them wedges, for the wedges of carpenters, of builders, of smiths and of the other arts are likewise fashioned unequal-sided in

every direction, beginning to penetrate from a sharper end and continually broadening, unequally in all their dimensions. Some call the same numbers ‘little wasps’, for the body of wasps too is very much of this kind, being pinched in at the middle and displaying the likeness spoken of. From this, it is likely, the sphekoma also has got its name, for wherever it draws tight it imitates the waist of the wasp. Others call the same numbers ‘little altars’, from a likeness that fits them, for the altars of ancient style, and especially the Ionic, have neither the breadth equal to the depth, nor both of these to the length, nor the base to the top, but are varied in their dimensions in every direction.

2.16.3

As between two extremes, then, the cube and the scalene, the one extended according to equality and the other according to inequality in every direction, there are intermediate the solid numbers called parallelepipedal, whose planes are heteromecic numbers, just as the planes of the cubes themselves were square numbers, as was shown.

Chapter 17

2.17.1

Once more, then, from the beginning: a number is called heteromecic when, if it is represented as a plane figure, the diagram proves to be four-sided and four-angled, yet the sides are not equal to one another, nor is the length equal to the breadth, but they differ by a unit, for example 2, 6, 12, 20, 30, 42 and those that follow; for if one represents them as plane figures, he will in every case do it thus: once 2 is 2, twice 3 is 6, thrice 4 is 12, and the succeeding ones analogously, four times 5, five times 6, six times 7, seven times 8 and so on without limit, provided only that the one side be greater than the other by a unit and by no other number. But if the sides differ from one another otherwise than by a unit, for instance by a dyad, a triad, a tetrad or so on in succession, as in twice 4 or thrice 6 or four times 8, or in whatever other way, such a number will no longer properly be called

heteromecic, but oblong. For the ancients of the school of Pythagoras and his successors saw the other and otherness in their root form in the dyad, and the same and sameness in the monad, as in two principles of all things; and these are found to differ from one another by a unit only, so that the other also is, in germ, other by a unit and by no other number; and for this reason, in ordinary usage too, among those who speak correctly 'the other' is said of two things and not of more. Moreover it was shown that every odd number receives its specific form from the monad, and every even number from the dyad. Hence we shall with good reason say that the odd partakes of the nature of the same, and the even of that of the other; for indeed by the heaping up of each there are produced, by nature and not by any ordinance of ours, from that of the odd numbers from the unit on without limit the nature of the squares, and from that of the even numbers from the dyad on without limit that of the heteromecic numbers.

2.17.3

It is therefore altogether necessary to think that the square, once more, partakes of the nature of the same, for its sides display towards one another the same ratio, alike, unvarying and resting in equality, and that the heteromecic number partakes of the nature of the other; for in the way in which the monad differs from the dyad, differing by a unit only, in that way also the sides of every heteromecic number vary from one another, the one differing from the other by a unit only. For example, when the continuous series of number from the unit is set out before me, I pick out and arrange the odd numbers by themselves in one row, and the even by themselves in another, and I get two rows such as these:

2.17.3

1	3	5	7	9	11	13	15	17	19	21	23	25	27
2	4	6	8	10	12	14	16	18	20	22	24	26	28

2.17.4

Now the beginning of the row of the odd numbers is the monad, which is of the same kind as it and possesses the nature of the same; wherefore neither, if it multiplies itself, in the plane or in the solid manner, is it made other, nor does it make any other number whatever depart from what that number was at the start, but it keeps it in the same state; and such a thing it is impossible to find in the case of any other number.

2.17.5

The other row is led by the dyad, which is of the same kind as it and is the originator of otherness; for whether it multiplies itself or another number, it causes a departure, as for example twice 2, twice 3.

2.17.6

But whenever there is eight times 8 taken twice or thrice,[17] such solid figures are called bricks, equal times equal times less; and if the heights added to the square are greater, such numbers are called beams, for example thrice 3 taken seven times or eight times or nine times, or any number of times whatever, provided only that it be in excess; and a beam is a number equal times equal times greater. The wedges, as we saw, were unequal times unequal times unequal, and the cubes equal times equal times equal.

2.17.7

Of the cubes themselves, those which, in addition to being equal times equal times equal, have also the property of always ending, at every multiplication, in the same number from which they began, are called spherical, and the same are also called recurrent, as indeed are the cube from the side 5 and that from the side 6; for by however many increases I increase each of these, it will always and in every case end in the same termination, that from 6 in 6 itself, and that from 5 in 5 itself. For example, five times 5 will end in 5, and so will this taken five times, and, if need be, this again taken five times; and on without limit no other ending will be found except 5, and from 6 in the same way 6, and no other. So that the unit also is potentially spherical and recurrent, for it undergoes, as is natural, the very same thing that

befalls spheres and circles; for each of those, where it begins, there also it ends, as it goes around in its circle and revolves. So too these numbers that have been mentioned, alone among the other equal times equal numbers, return to the same beginning from which they started, throughout all their increases. But if they advance in the plane manner, in two dimensions, they are called circular, as 1, 25, 36, from once 1 and five times 5 and six times 6; while if they have three dimensions, or are multiplied still further than these, they are called spherical solid numbers, as 1, 125, 216, or again 1, 625, 1296.

Chapter 18

2.18.1

Concerning solid numbers, then, this too is enough for the present. But since both the natural philosophers and those who take their start from the mathematical sciences say that the principles of all things are the same and the other, and since it has been shown that the same is the monad and the odd numbers that receive their specific form from it, and far more the squares that are constituted from these when they are heaped together, inasmuch as they partake of equality in their sides, while the other is the dyad and every even number that receives its specific form from it, and especially the heteromecic numbers that are constituted from these when they are heaped together, because they partake of the first inequality and otherness in the difference of their sides, it is most necessary to demonstrate this further point, how in both of these, as in principles and seeds, all the peculiar properties of number are potentially present beforehand, those of its species and its subdivisions, of all its relations, of the polygonal numbers and of the like.

2.18.2

But first we must distinguish wherein an oblong number differs from a heteromecic. For a heteromecic number, as was said before, is that which is produced by a number multiplying the number greater than itself by a unit, for example 6 by twice 3, 12 by thrice 4, while an

oblong number is that which is likewise itself produced by two differing numbers, differing however not by a unit, but by some greater number, for example twice 4, thrice 6, four times 8, and the like, which have in a manner run forward in length and overstepped the difference of a unit.

2.18.3

Accordingly, because the squares arise from certain numbers lengthened by their own length, having their length the same as their breadth, they might properly be called numbers 'of their own length' and 'of the same length', for example twice 2, thrice 3, four times 4 and those in succession; and if this is so, they are assuredly receptive of sameness and equality, and therefore are definite and limiting; for the equal and the same are such in one way, and that a definite one. And because the heteromecic numbers, on the other hand, are produced not by their own length but by the lengthening of another, they are for this reason both heteromecic and receptive of otherness, of unlimitedness and of indefiniteness.

2.18.4

In this way, then, all the things of number, and the things in the universe that have been brought to completion with reference to these, stand divided in two, are apportioned, and are seen to be contrary to one another; and rightly do the ancients, when they begin their account of nature, make the first division of their cosmogony in this way. Plato speaks by name of 'the nature of the same' and 'that of the other', and again of 'the being that is indivisible and always in the same state' and 'that which, in its turn, comes to be divisible'; and Philolaus says that it is necessary that 'all the things that are be either unlimited or limiting, or limiting and unlimited together', which last he rather agrees to be the case, namely that the universe is constituted out of limiting and unlimited things together, evidently after the image of number; for number too is as a whole composed of the monad and the dyad, of even and odd, which are indeed expressive

of equality and inequality, of sameness and otherness, of the limiting and the unlimited, of the definite and the indefinite.

Chapter 19

2.19.1

In order that we may also be clearly persuaded of what is being said, namely that the things that are were constituted out of warring and contrary things and with good reason admitted harmony (and harmony in every case arises out of contraries; for 'harmony is the unification of things much mingled and the agreement of things that think at variance'), let us set out in two parallel rows, lengthwise, no longer the even numbers from the dyad and the odd numbers from the unit by themselves, as a little while ago, but the numbers produced out of these very ones when they are heaped up with one another, squares from the odd, heteromecic numbers from the even; for if we fix our gaze upon the series of them set out we shall marvel at their friendship for one another and their helpfulness to one another in generating and completing the rest, so that we may with good reason conceive that in the nature of the universe also something of this kind is accomplished, from some such source as this, by the cosmic providence.

2.19.2

Let the two rows, then, be as follows, that of the squares from the unit, and that of the heteromecic numbers, itself likewise beginning from the dyad:

2.19.2

1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
2	6	12	20	30	42	56	72	90	110	132	156	182	210	240

2.19.3

In the first place, then, the first is the root multiple of the first, the second the sesquialter of the second, the third the sesquitertian of the third, the fourth the sesquiquartan of the fourth, then sesquiquintan

and sesquisextan, and so on analogously without limit; and their differences will advance according to the continuous series of number from the unit, a unit for the first pair, a dyad for the second, a triad for the third, and always thus. Next, if, to begin with, the second of the squares be paired and compared with the first of the heteromecic numbers, and the third with the second, and the fourth with the third, and the rest in sequence, they will preserve the same ratios unchanged as those before, but the differences, the same ones, will begin to advance no longer from the unit but from the dyad; and, by way of progression, in the former comparison the first will stand to the first in the first, the root form of the multiple, the second to the second in the sesquialter that is second from the root, the third to the third in the sesquitertian that is third from the root, and those that follow will advance in like manner.

2.19.4

Moreover, the squares have as their differences from one another only the odd numbers, and the heteromecic numbers the even. And if we set the first heteromecic number as a mean term between the two first squares, the second between the next, the third between those after them, and the fourth between those following, by these the relations will be seen to be still more orderly, in three terms; for as is the relation which 4 has to 2, so is 2 to the unit, and as 9 is to 6, in the sesquialter ratio, so is 6 to 4, and as 16 is to 12, so is 12 to 9, and so on in succession, the numbers and the ratios advancing in orderly fashion. For as the greater is to the mean term, so will the mean term be to the least, and not in the same ratio, but in one that always varies as it advances. And in all the couplings the product of the extremes is equal to the square of the mean term; and the extremes taken once, together with twice the mean term, will alternately in every case make a square; and, what is the most elegant of all these things, from the two put together there arises an orderly generation of triangular numbers, signifying that the nature of these is more primary than the principle of all, 1 and 2, and 2 and 4, and 4 and 6, and 6 and 9, and 9

and 12, and 12 and 16, and 16 and 20, and always in this way the triangular numbers that generate the polygonal numbers come to be in orderly fashion.

Chapter 20

2.20.1

Furthermore, every square, when it takes on its own side in addition, becomes heteromecic, or, by Zeus, when it has its own side taken away. Thus the other too is conceived both as more and as less than the same, since it is brought about by addition and by subtraction, just as the two species of the unequal also, the greater and the less, take their origin by addition or subtraction applied to the equal.

2.20.2

This too is sufficient evidence that both the forms partake of sameness and otherness, of otherness indefinitely, of sameness definitely; generically the monad and the dyad do so, and in a subordinate way the odd partakes of sameness, because it is of the same kind as the monad, and the even of otherness, because it is of the same kind as the dyad.

2.20.3

And it is plainer still that the square, because it is a sum of odd numbers, is akin to sameness, and the heteromecic number, because it is a sum of even numbers, to otherness. For indeed, as being friendly to one another, these two forms in the two rows give one another a share, by turns, of the same differences, if not also of the same ratios, and conversely of the same ratios, if not also of the same differences; for what lies between 4 and 2 in the double ratio, this lies between 6 and 4 in a superparticular ratio, and again what lies between 9 and 6 in the sesquialter ratio, this lies between 12 and 9 in the sesquitertian, and always thus; and what is the same in quality is other in quantity, and contrariwise what is the same in quantity is other in quality.

2.20.4

And again, there is the fact that necessarily, throughout all the relations, the same difference of the two terms will be called a part in a manner varying by a unit, being a half of the one and a third of the other, or a third of the one and a fourth of the other, or otherwise a fourth of the one and a fifth of the other, and so on in succession.

2.20.5

But what will most of all confirm that the odd is in the highest degree the cause of sameness, and never the even, is this, which must be pointed out: in every series set out in proportion from the unit, for example in the double ratio 1, 2, 4, 8, 16, 32, 64, 128, 256, and in the triple 1, 3, 9, 27, 81, 243, 729, 2187, and as far as you please, you will find that of necessity all those in the odd places are squares, and no others besides by any device, and that none in an even place is a square. Moreover the equal times equal times equal numbers, all of them, that is, the cubes, which are extended in three dimensions and seem to partake of sameness in a still higher degree, are the work of the odd numbers, and not of the even, 1 and 8 and 27 and 64 and 125 and 216 and those that advance analogously, and that by a simple and unvarying method. For when the successive odd numbers from the unit are set out without limit, observe as follows: the first makes the potential cube, the two after it, added together, the second, the three following upon these the third, the four continuous with these the fourth, the five next in order after these the fifth, and the six following the sixth, and so on for ever.

Chapter 21

2.21.1

After this it would be the right moment to add the treatment of proportions, which is most necessary for the investigations of nature and for the theorems of music, of sphaeric and of lines, and not least for the reading of the ancients, and so to put to the Introduction to Arithmetic the end that is fitting and at the same time of the most suitable measure.

2.21.2

A proportion, then, is in the proper sense the bringing together into one of two or more ratios, but more generally of two or more relations, even if they are not subjected to the same ratio, but to a difference or to something else.

2.21.3

Now a ratio is the relation of two terms to one another, and the composition of such is the proportion, so that this is combined in three terms at the least, though it can also advance in more, according to the same interval or according to the same ratio. For example, the ratio of 1 to 2, there being two terms, is one ratio, the double; but there is also another like ratio of 2 to 4; therefore 1, 2, 4 is a proportion, for it is a bringing together of ratios, or of three terms regarded in the same ratio to one another.

2.21.4

And in more numerous and longer series set out the same thing can be observed; for let a fourth term, 8, be attached after 4, again in a like relation, for it is double, and again after 8, 16, and always thus.

2.21.5

If, then, the same term, always one and unchanged, answers to those on either side of it, to the greater as consequent and to the less as antecedent, such a proportion is called continuous, for example 1, 2, 4 in respect of quality; for as 4 is to 2, so is the same 2 to 1, and conversely as 1 is to 2, so is the same 2 to 4; and in respect of quantity, for example 1, 2, 3; for by as much as 3 exceeds 2, by so much does 2 itself also exceed 1, and contrariwise, by as much as 1 falls short of 2, by so much does 2 itself also fall short of 3.

2.21.6

But if one term answers to the less, becoming antecedent and greater, and another, and not the same, to the greater, becoming consequent and less, such a mean and proportion is called no longer continuous but disjunct; for example, in respect of quality, 1, 2, 4, 8; for as 2 is to 1, so is 8 to 4, and conversely as 1 is to 2, so is 4 to 8, and alternately as 1

is to 4, so is 2 to 8, or as 4 is to 1, so is 8 to 2; and in respect of quantity thus, 1, 2, 3, 4; for by as much as 1 falls short of 2, by so much also does 3 fall short of 4, or by as much as 4 exceeds 3, by so much also does 2 exceed 1, or again in mixed order, by as much as 3 exceeds 1, by so much does 4 exceed 2, or by as much as 1 falls short of 3, by so much does 2 fall short of 4.

Chapter 22

2.22.1

Now the proportions that are first and acknowledged by all the ancients, by Pythagoras and Plato and Aristotle, are the three very first, the arithmetic, the geometric and the harmonic; and there are three others subcontrary to these, which have not obtained names of their own, but are called more generally the fourth, fifth and sixth means; after which the more recent writers find four others as well, filling out the number ten, which in the opinion of the Pythagoreans is the most perfect, the number according to which the ten relations too were seen by us a little while ago to take their quantity, and the ten categories, as they are called, and the divisions and relations of the extremities of our hands and feet, and countless other things, which we shall see in their proper place elsewhere.

2.22.2

But now we must treat of the proportions systematically from the beginning, and first of that which, by quantity, makes the comparison of the terms akin to one another and binds them together, the one which, in respect of the difference of the terms from one another, is equal in quantity; and this would be the arithmetic, for quantity was earlier assigned to it as its own.

2.22.3

What then is the reason that we treat of this first and not of another? Surely it is plain that nature too displays it before the others; for in the natural series of simple number set out from the unit, with nothing passed over or withdrawn, the definition of this one alone is

preserved; but in what went before, too, we demonstrated by reasoning that the Introduction to Arithmetic itself is prior to all the others, since it abolishes them together with itself but is not abolished together with them, and is implied together with them but does not imply them together with itself; so that with good reason the mean that bears the same name as arithmetic will not unreasonably take precedence of the means that bear the names of those others, the geometric and the harmonic; for it is most plain that it will much more take precedence of the subcontraries, of which these are the leaders. Since, then, the arithmetic is by nature the very first and most rightfully the leader, let it obtain an articulate treatment from us also before the others.

Chapter 23

2.23.1

An arithmetic mean exists, then, whenever, three or more terms being set out or conceived in succession to one another, the same difference in quantity is found to exist between the successive terms, while the same ratio does not arise among the terms in relation to one another, for example 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13; for in this natural series of number, when it is examined continuously and with no term passed over, every term whatsoever, ranged between two others, is found to preserve the arithmetic mean in relation to them; for its differences from those ranged on either side are equal, yet the same ratio is not also preserved among them.

2.23.2

And we know that in such a series there arises both a continuous and a disjunct mean; for if the same mean term answers as both antecedent and consequent to those on either side, it would be continuous, but if another term together with it, the mean becomes disjunct.

2.23.3

If, then, cutting off from this series any three adjacent terms whatsoever, we examine them according to the continuous form, or

four or more according to the disjunct, the unit would be the difference of them all, but the ratios different in every case; and if we take terms not adjacent but at intervals, yet with an equal omission, and again either three or more, then, if one term be omitted at each setting of a term, the dyad will be the difference of them all, and again in three terms the mean will be continuous and in more disjunct; and if those omitted be two, the triad will in every case be the difference in all, continuous and disjunct alike, and if three, the tetrad, and if four, the pentad, and so on to any extent whatever.

2.23.4

Such a mean, therefore, partakes of equal quantity in its differences, but no longer of equal quality; for this reason it is arithmetic. But if, conversely, it partook of like quality but not of quantity, it would be geometric instead of arithmetic.

2.23.5

It is a property of this mean, and one that belongs to no other, that the mean term is either half of the sum of the extremes or equal to it, whether it be examined as continuous or as disjunct or alternately; for either the mean term with itself or the mean terms with one another are equal to the sum of the extremes.

2.23.6

It has yet another property as well: the ratio which each term has to itself, this the differences have to the differences, that is, they are in equality. Further, the most elegant property, and one that has escaped the notice of most: the product of the extremes, when compared with the square of the mean term, is found to be less than it by the product of the differences, whether these be units or dyads or triads or tetrads or any number whatsoever. And a fourth, which all the earlier writers also noted: the ratios in the lesser terms, when compared with those in the greater, are greater; and it will be shown that in the harmonic, on the contrary, those in the greater terms are greater and those in the lesser less. For this reason the harmonic mean is subcontrary to the arithmetic, and the geometric is a middle ground between them as

between extremes, having the ratios in the greater terms and those in the lesser equal to one another; and the equal was seen by us to lie in the middle between the greater and the less. This much we have to say concerning the arithmetic mean.

Chapter 24

2.24.1

The geometric mean, which comes next in order after this one, is alone called a proportion in the proper sense, because the terms in it are seen to stand to one another in the same ratio. It exists whenever, of three terms or more, as the greatest is to the one beneath it, so is this to the one below it, and, if there are more terms, so again is this last to the one beneath it, while they nevertheless do not differ from one another by the same quantity, but by the same quality of ratio, the opposite of what was seen in the case of the arithmetic.

2.24.2

For instance, by way of example, let there be set out the numbers that advance from the unit in the double ratio, 1, 2, 4, 8, 16, 32, 64 and so on to infinity, or in the triple, 1, 3, 9, 27, 81, 243 and so on in order, or in the quadruple, or in a manner similar to those set out; for in each of these rows three adjacent terms, or four, or as many as may be taken, will yield the geometric proportion to one another: as the first is to the one beneath it, so is that one to the one beneath it, and again that one to the one still beneath it, and this as far as one wishes, and in mixed order as well. For example, 2, 4, 8; for the ratio which 8 has to 4, this 4 also has to 2, and inversely; yet they do not have an equal quantity between one another. Or again 2, 4, 8, 16; for not only does 16 have to 8 the same ratio as the preceding terms, though not the same difference, but in mixed order also it preserves the like relation: as 16 is to 4, so also is 8 to 2; and in reverse, as 2 is to 8, so also is 4 to 16; and disjunctly, as 2 is to 4, so is 8 to 16; and conversely and in the disjunct form, as 16 is to 8, so also is 4 to 2; for it has the double ratio.

2.24.3

The geometric mean has a property which none of the others has, namely that the differences of the terms are in the same ratio to one another as that in which the terms stand to those adjacent to them, the greater to the less, and the differences taken inversely are as the terms taken inversely. There is yet another peculiar property: the greater terms have as their difference from the lesser the lesser terms themselves, and in the same way difference differs from difference by the lesser difference itself, when the terms are set out in the double ratio; in the triple ratio both the terms and the differences will have as their difference twice the one beneath them, in the quadruple three times, in the quintuple four times, and so on throughout.

2.24.4

Geometric proportions arise not only among multiples, but also among all the species of superparticulars, and among superpartients and mixed ratios, and the distinctive property of this mean is preserved in all of them: in the continuous proportions the product of the extremes is equal to the square of the mean term, and in those that are disjunct, or even in a greater number of terms, though they be not continuous, provided they are even in number, the product of the extremes is equal to the product of the means.

2.24.5

As an illustration that in all the relations, in multiples of every kind, superparticulars of every kind, superpartients of every kind and mixed ratios of every kind, the peculiar property of this proportion is preserved, let that be sufficient and adequate for us in which, starting from equality, we fashioned by the three rules all the species of the unequal out of one another, both when the terms are set in direct order and when they are reversed; for each such formation and series is a geometric proportion, embracing all the properties mentioned and a fourth besides, that it maintains the same ratio both in the greater terms and in the lesser. Moreover, if we set out the common row of heteromecic numbers and squares, containing alternately, one by one, the numbers of both kinds, and then, cutting them off by

threes from the unit onward, examine them, always making the last of the former group the beginning of the following, we shall find, starting from the multiple relation, that is, the double, all the superparticulars appearing in order: sesquialter, then sesquitertian, then sesquiquartan, and so on.

2.24.6

It would be most opportune, now that we have come to this point, to mention a corollary that is of use to us for a certain Platonic theorem; for plane numbers are in every case held together by one mean, but solids by two means set in proportion. For between two consecutive squares only one mean term is found that preserves the geometric proportion, antecedent to the lesser and consequent to the greater, and never more than one; we therefore observe two intervals, from the mean term itself to each of the extremes, in a relation of like ratios.

2.24.7

Again, between two consecutive cubes only two mean terms in proportion are found, according to the geometric proportion, and never more; there are therefore three intervals, one that between the mean terms in relation to each other, and two those between the extremes and the mean terms on either side.

2.24.8

Thus solid figures are said to be extended in three dimensions, and plane figures in two; for example, 1 and 4 are plane numbers, and 2 is their mean proportional; or again 4 and 9, two squares, and their mean proportional is 6, which is held by the greater and holds the lesser in the same ratio as that in which difference also holds difference.

2.24.9

The cause of this is that the sides of the two squares, one belonging to each, both together generated this very 6. But in cubes, for example 8 and 27, there is no longer one mean, but two are found, 12 and 18, which set themselves and the terms in the same ratio as that in which

their differences also stand to one another; and the cause of this too is that the two means are a combination of the sides of the cubes intermingled, twice 2 three times and thrice 3 twice.

2.24.10

In general, then, if a square takes a square, that is, multiplies it, it always makes a square, but if a square multiplies a heteromecic number,[18] a square is never produced; and if a cube multiplies a cube, a cube will always result, but if a heteromecic number multiplies a cube, never a cube; just as, of course, if an even number multiplies an even, an even number is always generated, and if an odd multiplies an odd, always an odd, but if an odd multiplies an even or an even an odd, the product will always be even and never odd.

2.24.11

These matters will receive their proper elucidation in the reading of Plato, at the passage in the Republic on the so-called marriage, which is introduced in the person of the Muses. Let us therefore pass on to the third proportion, the one called harmonic, and analyse it.

Chapter 25

2.25.1

Now the mean that is third in order, called the harmonic, exists whenever among three terms the mean term is observed, in relation to the extremes, neither as standing in the same ratio, antecedent to the one and consequent to the other, as in the geometric, nor in equal intervals but with otherness of ratios, as in the arithmetic, but when, as the greatest term is to the least, so also is the difference of the greatest from the mean term to the difference of the mean term from the least, for example 3, 4, 6, or again 2, 3, 6;[19] for 6 exceeds 4 by its own third part, since 2 is a third of 6, and 3 falls short of 4 by its own third part, since the third of 3 is a unit. For in the former example the extremes are in the double ratio, and their differences from the mean term are again in the same double ratio to one another, while in the second each is in the triple.

2.25.2

It has a peculiar property subcontrary, as we said before, to that of the arithmetic mean; for there the ratios were greater among the lesser terms and less among the greater, but here, on the contrary, those among the greater terms are greater and those among the lesser are less, so that in the geometric, as in a mean between the two, the equality of the ratios on either side may with good reason be observed, being a middle ground between the greater and the less.

2.25.3

Further, in the arithmetic the mean term is seen to be greater and less than those on either side by the same part of itself, but greater or less by one part and another of those terms themselves; in this harmonic mean it is the contrary, for the mean term is greater and less than those on either side by one part and another of itself, but in every case by the same part of those very terms on either side, for it is either by a half of each or by a third of both. The geometric, as on the middle ground between the two, has this[20] neither in the mean term alone nor in the extremes alone, but in both, in mean and extreme.

2.25.4

Further, the harmonic has as an attribute of its own that the extremes, added together and multiplied by the mean term, make double the product of their multiplication by one another.

2.25.5

A mean of this kind was called harmonic because the arithmetic was distinguished by quantity, exhibiting equality in this respect in the interval of the terms from one another, and the geometric by quality, yielding relations that are alike in respect of the quality of the terms towards one another, while this one, in respect of what is in some way relative to another, presents itself in forms now of one kind and now of another; for it is neither in the terms alone nor in the differences alone, but partly in the terms and partly in the differences; for as the greatest is to the least, so also is the difference of the greatest from

the mean term next to it to that of the least from the same mean term, and inversely.

Chapter 26

2.26.1

Now the relative we recognized, in the division of being set forth above, as proper to the theory of harmony; and besides, the musical ratios of the concords in harmony are found rather in this mean. Most elementary is the fourth, in the sesquitertian ratio, as 4 to 3, term to term in the example of the double, or difference to difference in that of the triple, for these are the differences of 6 from 2 or again of 6 from 3. Directly after this comes the fifth, which is sesquialter, that of 3 to 2 or again of 6 to 4, term to term. Then the combination of both of these, the sesquialter and the sesquitertian, is the octave, lying next in order to them, which is in the double ratio, as 6 to 3 in both examples, term to term; or the one after this, the octave and fifth together, which preserves the triple ratio of both together, being a combination of double and sesquialter at once, like that of 6 to 2, term to term, in the example of the triple, and again that of difference to difference in the same example, and, in the example of the double, that of the greatest term to the difference between it and the mean term, or of the difference of the extremes to the difference of the lesser terms. Last and greatest concord is the so-called double octave, as it were twice double, which is in the quadruple ratio, as the mean term of the proportion in doubles is to the difference of the lesser terms, or as the difference of the extremes of the proportion in triples is to that of the lesser terms.

2.26.2

Some, however, following Philolaus, think that it is called harmonic from its attending upon every geometric harmony; and geometric harmony is what they call the cube, from its having been harmonized in its three dimensions, equal times equal times equal. For in every cube this mean is mirrored: the sides of every cube are 12, its angles 8

and its faces 6; 8, therefore, is the mean of 6 and 12 according to the harmonic proportion. For as the extremes are to one another, so is the difference of the greatest from the mean term to the difference of the mean term from the least; and again, the mean term is greater than the lesser by one part of itself and less than the greater by another, yet it is both greater and less by one and the same part of the extremes themselves; and, in another way, the extremes added together and multiplied by the mean term make double the product of the extremes by one another. And the fourth is the ratio of 8 to 6, for it is sesquitertian; the fifth that of 12 to 8, for it is sesquialter; the octave, which is the combination of the two, that of 12 to 6, for it is double; the octave and fifth together, which is triple, is the difference of the extremes in relation to that of the lesser terms; and the double octave is the mean term in relation to the difference between itself and the lesser. Most appropriately, therefore, has it been named harmonic.

Chapter 27

2.27.1

Just as in the division of the musical canon, when a single string is stretched or a single length of pipe is laid out, the extremities remaining immovable while the middle is shifted, in the pipe by means of the finger-holes and in the string by means of the bridge, the means already mentioned, arithmetic, geometric and harmonic, can be produced in one way after another, so that they would with good reason and in the truest sense of the word be called means, brought about as they are in different ways through the displacement and shifting of the mean term; so also with two numerical terms, whether both odd or both even, it is reasonable and at the same time possible, while they remain as they are and are not shifted, for a mean to be inserted that fits according to each of the three: according to the arithmetic, one that exceeds and is exceeded by an equal amount; according to the geometric, one that differs by a like ratio; and

according to the harmonic, one that is greater and less by the same part of the same extremes.

2.27.2

Let there first be set before us, then, two even terms, between which we must inquire how the three means would be placed and what they are, and let them be 10 and 40.

2.27.3

First, then, I fit in the arithmetic mean, and it is 25, and all its consequences are preserved here too: for as each term is to itself, so also is difference to difference, and they are therefore in equality; and by as much as the greater exceeds the mean term, by so much does this in turn surpass the lesser; and the sum of the extremes is double the mean term; and the ratio of the lesser terms is greater than that of the greater; and the product of the extremes is less than the square of the mean term by the square of the differences; and the mean term is both greater and less than the extremes by the same part of itself, but by one part and another when the part is regarded in the extremes.

2.27.4

But if I insert the mean 20 into the given even terms, the properties of the geometric emerge, and those of the arithmetic are utterly lost; for as the greater is to the mean term, such also is the mean term to the smaller, and the product of the extremes is equal to the square of the mean term, and the differences are seen to be in the same ratio to one another as that in which the terms also stand, and the sameness of the part in excess and deficiency is neither in the extremes alone nor in the mean term alone, but in the mean term and one of the extremes alternately, and there is an equal ratio both in the greater terms and in the lesser.

2.27.5

And if I take 16 instead as the mean term, again the properties of the two former means pass out of sight, and those of the harmonic come to light, holding good in relation to the same two even terms; for as the greater is to the least, so is the difference of the greater terms to

that of the lesser; and by as many parts, regarded in the greater term itself, as the mean term is less than the greater, by so many, regarded in the lesser itself, is the same term greater than the lesser; and the ratio in the greater terms is greater, and that in the lesser terms less, which is the case in no other; and the extremes added together and multiplied by the mean term make double the product of the extremes.

2.27.6

But if the two extreme terms are set out not even but odd, for example 5, 45, the same term, 25, will make the arithmetic mean; and the cause of this is that on either side of it the terms have gone above and below by an equal number, keeping the same quantity of difference in relation to it. And 15, set in its place, yields the geometric, being the triple of the one and the subtriple of the other; and 9, taking over the part of mean term, yields the harmonic, for by the parts by which it is greater than the lesser, four fifths of the lesser itself, by these it is less than the greater, regarded in the greater itself, for it is by four fifths; and on applying all the properties previously mentioned you will find them in agreement.

2.27.7

A method by which you might skilfully fashion the terms shown above according to the three proportions shall be for you as follows. In the case of both pairs of terms that were taken in hand, the odd and the even, you will find the arithmetic mean thus: add the extremes and set half of them as the mean term, or, cutting in two the excess of the greater over the lesser and adding it to the lesser, you will have the mean term. The geometric you will find by finding the square root of the oblong contained by the extremes and making it the mean term, or, having observed the ratio which the terms have to one another, cut this in half and make the mean term, for example the double in the case of the quadruple. And for the harmonic, the difference of the extremes must be multiplied by the lesser, and the product divided by

the sum of the extremes, and then the quotient must be added to the lesser, and the result will be the harmonic mean.

Chapter 28

2.28.1

So much, then, for the three proportions commonly spoken of among the ancients, which we have purposely set out more clearly and at greater length, because it was possible to meet with them often and in more varied forms in what one reads; but those that come next must be treated summarily, since they are not much current among the ancients, but are taken up solely for our own acquaintance with them and for what may be called the completeness of the reckoning.

2.28.2

They are set forth by us in order according to their subcontrariety to the three archetypes already described, since they are in fact fashioned out of these very ones, and they obtain a like order.

2.28.3

The fourth, which is also called subcontrary because it is opposed to the harmonic and stands in reciprocal relation to it, exists when, in three terms, as the greatest is to the least, so the difference of the lesser terms is to that of the greater, for example 3, 5, 6, for the things compared are seen to be in the double ratio.

2.28.3

It is plain in what respects it is opposed to the harmonic: for while both have the same extremes, and in the double ratio at that, in the one before this the excess of the greater terms preserved the same ratio to that of the lesser, whereas in this, inversely, that of the lesser does so to that of the greater. And this must be known as its peculiar property, that the product of the greater and the mean term turns out double the product of the mean term and the least; for six times 5 is double five times 3.

2.28.4

The two means, the fifth and the sixth, were both fashioned after the geometric, and they differ from one another as follows: the fifth exists when, in three terms, as the mean term is to the least, so also the difference of these same terms is to that of the greatest from the mean term, for example 2, 4, 5; for 4 is double 2, the mean term double the least, and 2 is double 1, the difference of the least terms as against the difference of the greatest. What makes it subcontrary to the geometric is this, that in the latter, as the mean term is to the lesser, so the excess of the greater over the mean term is to that of the mean term over the lesser, whereas in this, inversely, that of the lesser is to that of the greater. Nevertheless this one too has a peculiar property, that the product of the greatest and the mean term is double the product of the greatest and the least; for five times 4 is double five times 2.

2.28.5

The sixth comes about when, in three terms, as the greatest is to the mean term, so the excess of the mean term over the least is to that of the greatest over the mean term, for example 1, 4, 6, for both are in the sesquialter ratio. The cause of its subcontrariety to the geometric is similar in this case as well, for here too the likeness of the ratios is reversed, as in the fifth.

2.28.6

These, then, are the six means commonly spoken of among our predecessors: three, the prototypes, which lasted from Pythagoras onward down to Aristotle and Plato, and three others, subcontrary to those, which came into use among the writers of commentaries and the sectarians who came after them. But certain persons, by shifting the terms and differences of these, have devised four others besides, which do not much appear in the writings of the ancients, but have been worked out with a somewhat over-curious subtlety; and these nevertheless, that we may not seem to be ignorant of them, we must run over in some such way as this.

2.28.7

The first of them, but the seventh in the ordering of them all, exists when, as the greatest is to the least, so also the difference of these same terms is to that of the lesser, for example 6, 8, 9; for the ratio of each is seen, on comparison, to be sesquialter.

2.28.8

The eighth mean, which is the second of these, comes about when, as the greatest is to the least, so the difference of the extremes is to the difference of the greater terms, for example 6, 7, 9; for this one also has its two ratios sesquialter.

2.28.9

The ninth in the ordering of them all, but the third in the number of those that have been discovered in addition, exists when, there being three terms, the ratio which the mean term has to the least is the same as that which the excess of the extremes has to that of the least terms, as 4, 6, 7.

2.28.10

The one that comes after all the rest, the tenth in sum but the fourth in the exposition of the more recent ones, is seen when, in three terms, as the mean term is to the least, so also the difference of the extremes is to the difference of the greater terms, for example 3, 5, 8; for the ratio in each pair is superbipartient.

2.28.11

By way of summary, then, let the terms of the ten proportions be set out in a single illustration, so that they may be easily taken in at one view:

2.28.11

Of the first	1	2	3
Of the second	1	2	4
Of the third	3	4	6
Of the fourth	3	5	6
Of the fifth	2	4	5
Of the sixth	1	4	6
Of the seventh	6	8	9
Of the eighth	6	7	9
Of the ninth	4	6	7
Of the tenth	3	5	8

Chapter 29

2.29.1

It remains for me to set forth briefly also the most perfect mean, which is extended in three dimensions and embraces all the others, and is most useful for all progress in music and in the study of nature; for this alone, in comparison with the others, would properly and truly be called harmony, since it is not plane, nor bound together by one mean only, but by two, so that it may thus be extended in three dimensions, just as it was made clear a little while ago that the cube is harmony.

2.29.2

Whenever, then, there being two extreme terms, both of them extended in three dimensions, whether equal times equal times equal, so as to be a cube, or equal times equal times unequal, so as to be either beams or bricks, or unequal times unequal times unequal, so as to be scalene, two other terms are found between them which preserve the same ratios to the extremes alternately and in mixed order, so that, whichever of them preserves the harmonic proportion, the remaining one completes the arithmetic, it is necessary, when the four are so disposed, that the geometric should appear, set in interwoven fashion over against both the means: as the greatest is to the third from it, so the second, the one next below it, is to the fourth;

for such a relation makes the product of the mean terms equal to that of the extremes. And again, if the greatest be shown to stand at as great a difference from the one next below it as this latter itself stands from the least, such a comparison proves arithmetic, and the sum of the extremes is double the mean term; and if the third from the greatest exceeds and is exceeded by the same part of the extremes themselves, it is harmonic, and the product of the mean term and the sum of the extremes is double the product of the extremes.

2.29.3

Let an example of it be the following: 6, 8, 9, 12. Now 6 is a scalene number, from once 2 thrice, and 12 is from twice 2 thrice, these being multiplied out in continuous succession; and of the mean terms the lesser is from once 2 four times, and the greater from once 3 thrice; and the extremes are solid and extended in three dimensions, and the means are of the same kind with them; and according to the geometric proportion, as 12 is to 8, so 9 is to 6; according to the arithmetic, by as much as 12 exceeds 9, by so much does 9 also exceed 6; and according to the harmonic, by the part by which 8 exceeds 6, the part being regarded in 6 itself, by that same part it is exceeded by 12, the part being regarded in 12 itself.

2.29.4

Moreover, 8 to 6, or 12 to 9, is the fourth, in the sesquitertian ratio; 9 to 6, or 12 to 8, is the fifth, in the sesquialter; and 12 to 6 is the octave, in the double. Finally, 9 to 8 is the interval of a tone, in the sesquioctave, which is the common measure of all the ratios in music, since it is also the more familiar, because it is in fact the difference of the first and most elementary concords from one another.

2.29.5

And concerning the things that appear in numbers and are their attributes, let this much suffice, as for a first introduction.

Notes

[1] The text as printed has no subject for δοκοῦντι; several manuscripts add τοί, 'those concerned with the mathematical sciences seem to me to have discerned well'.

[2] Hoche prints εἰ γένοιτο ἄν οὕτως; the translation follows καὶ γένοιτο ἄν, the reading of two manuscripts.

[3] Hoche prints μονάδος σύστημα; the translation follows μονάδων, the reading of three manuscripts and the wording of the definition at 1.7.1.

[4] The text has ἀπὸ μονάδος ('from the unit'), but the series that follows begins with 6, twice the triad; ἀπὸ τριάδος is said of the same odd numbers at 1.10.6.

[5] The Greek has only τὸ ὑπὸ ἴσον τῷ ἀπὸ ἢ τῷ ὑπό ('the product equal to the square or to the product'); the terms are supplied, here and at 1.10.10 and 2.19.4, from the full statement at 1.8.14.

[6] Hoche prints κατὰ τὴν ἑαυτῶν ποσότητα ('their own'); the translation follows ἑαυτοῦ, the original reading of one manuscript, which the next clause (μὴ τῇ ἑαυτοῦ) presupposes.

[7] The label of the diagram is μῆκος ('length') in two manuscripts and πλάτος ('breadth') in the rest; the description that follows (1.19.11) speaks of breadth and depth.

[8] The text as printed has ἑνάδος (so two manuscripts; others ἐνάδος, ἐννεάδος), which as accented means 'unit'; nine is meant, since the third rows begin from 9. A like form stands for nine at 1.19.13 and 1.22.4.

[9] The text has ἀπὸ τοῦ τρίτου ('from the third'); the series 3, 6, 9 … shows that the number three is meant.

[10] The series stops at 54, with six terms against the eight quadruples set beside it; one manuscript adds ξγ, οβ (63, 72).

[11] Strictly the last term is the same only in the two derived series (9, 15, 25 and 4, 10, 25), not in the series from which both come (9, 6, 4); several manuscripts add a fuller statement to this effect.

[12] What follows is not Nicomachus: the three diagrams, with their headings, are illustrations which Hoche prints from the Göttingen manuscript (the references to §§ 8–10 are his own), and the closing line is the subscription of one manuscript.

[13] The figures of the manuscripts, here and at each of the steps that follow (the successive triangular numbers set out in units), are not reproduced.

[14] The figure of the manuscripts (the square numbers set out in units) is not reproduced.

[15] The figures of the manuscripts, here and at the two places that follow (the pentagonal numbers 1, 5 and 12 set out in units), are not reproduced.

[16] The words καὶ κατὰ πλάτος ('and in breadth') have been suspected: it is down the columns that successive numbers differ by a triangular number (1, 3, 6, 10 ... from column to column), whereas along a row the differences are not triangles.

[17] The text returns abruptly, with no subject expressed, to the solid numbers left at 2.16.3; nothing is marked as lost, but the transition is wanting.

[18] The text adds here ἢ ἑτερομήκης τετράγωνον ('or a heteromecic number a square') and, after 'multiplies a cube' below, ἢ κύβος ἑτερομήκη ('or a cube a heteromecic number'); Hoche brackets both, though with hesitation, and they are not translated.

[19] The explanation that follows concerns the first example, 3, 4, 6, and a property not yet stated: that the greater term exceeds the mean, and the lesser falls short of it, by the same part of itself. One manuscript inserts a clause to that effect after the examples.

[20] The Greek has no verb; what is understood is the sameness of the part, as the fuller statement at 2.27.4 shows.